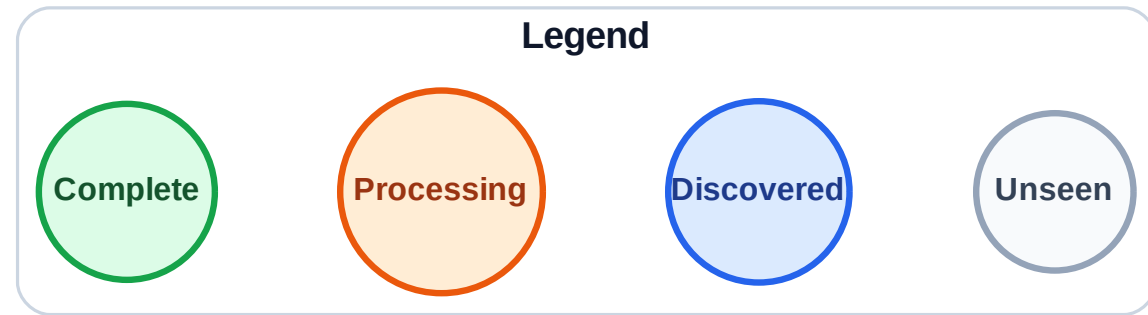
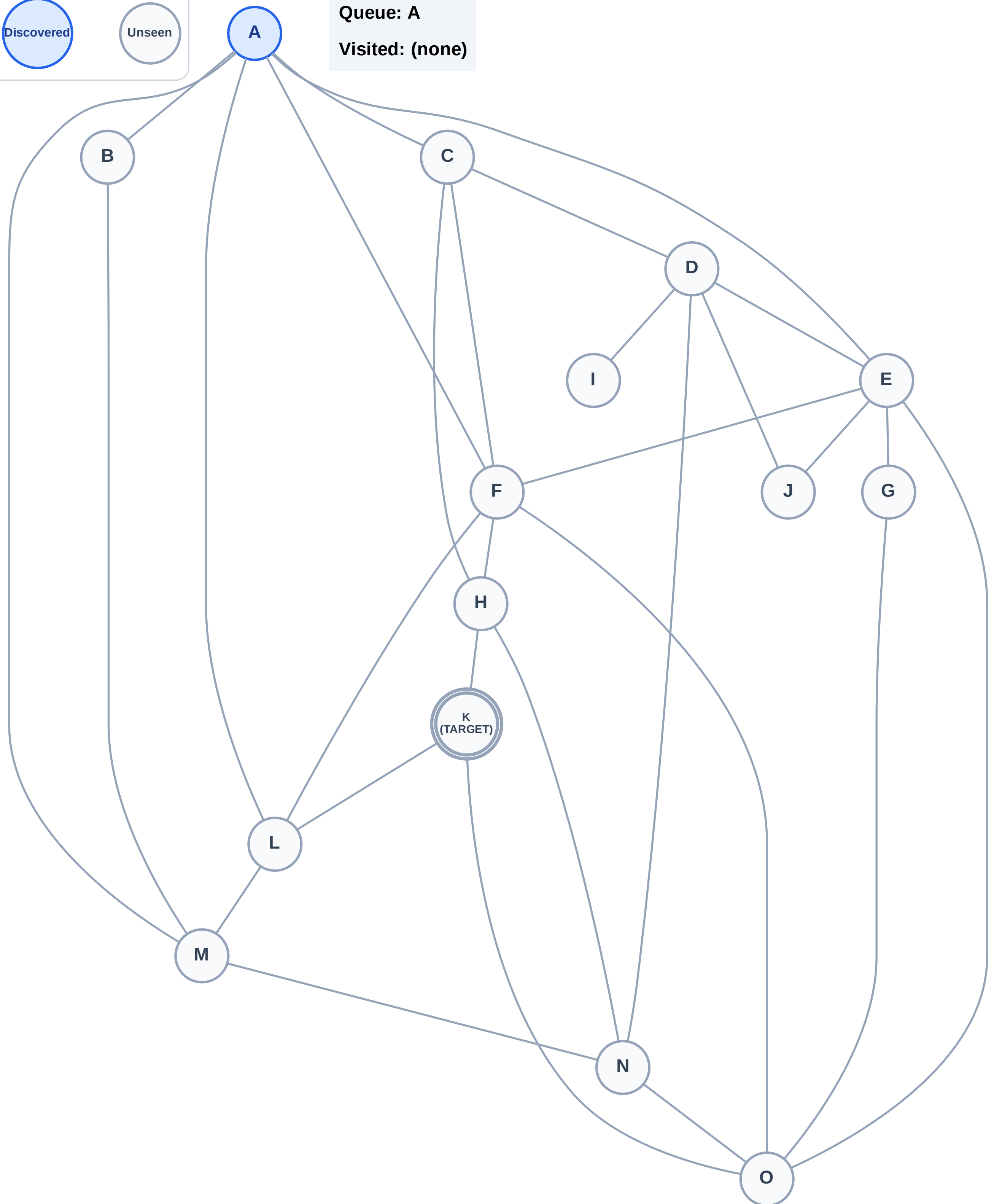


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

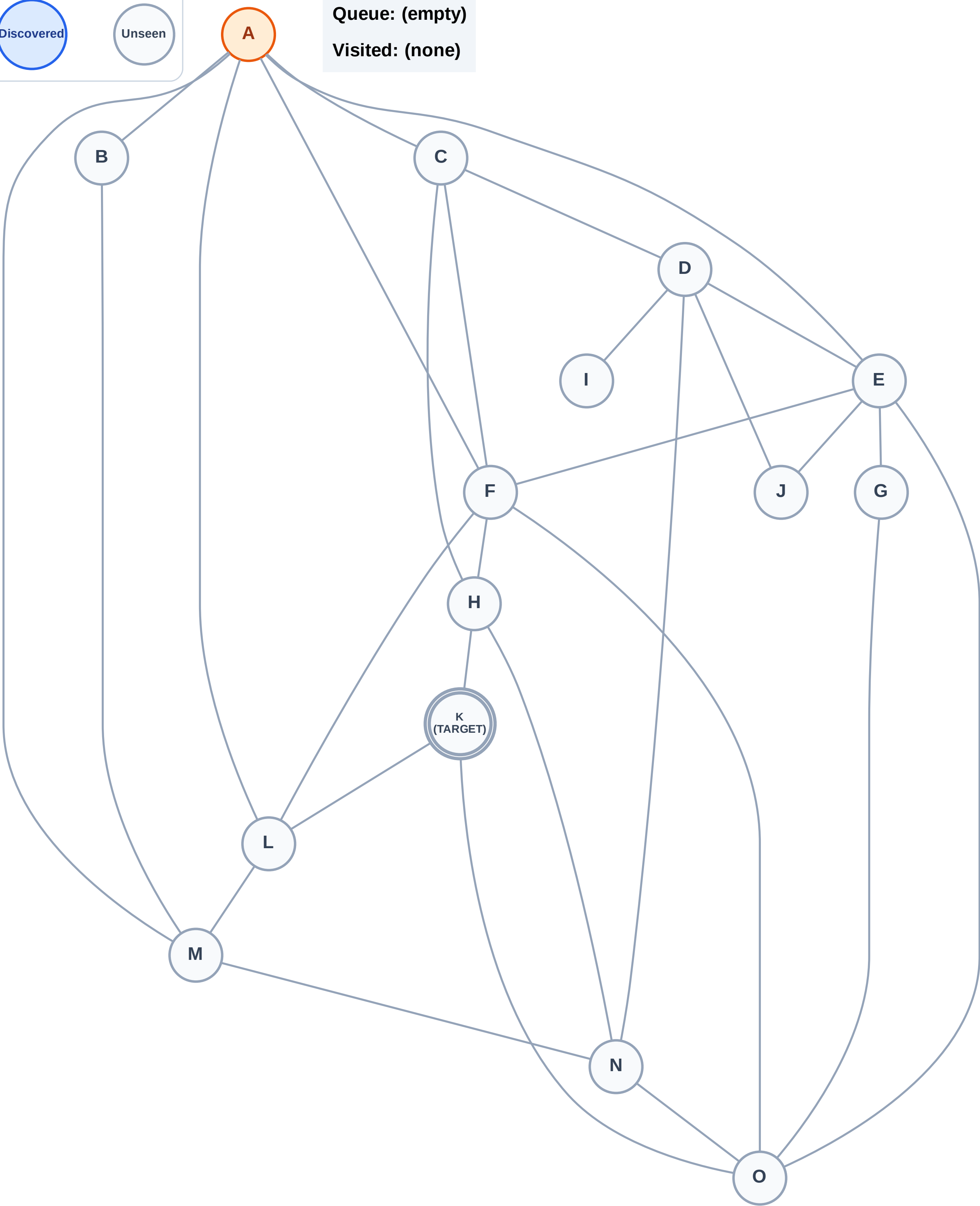
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



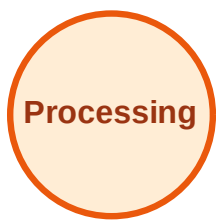
BFS Traversal

Step 3: Discover B, C, E, F, L, M from A

Legend



Complete



Processing



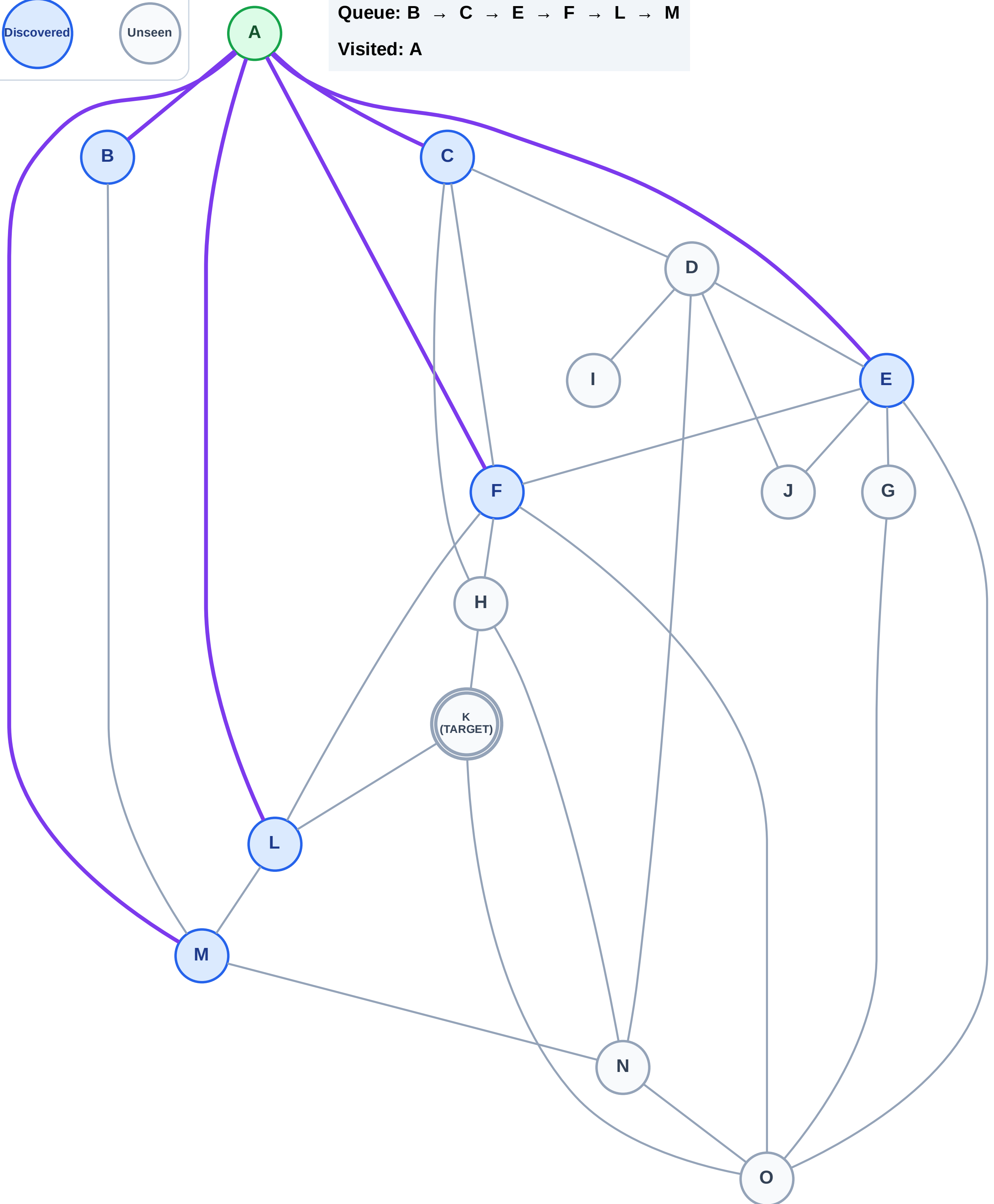
Discovered



Unseen

Queue: B → C → E → F → L → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

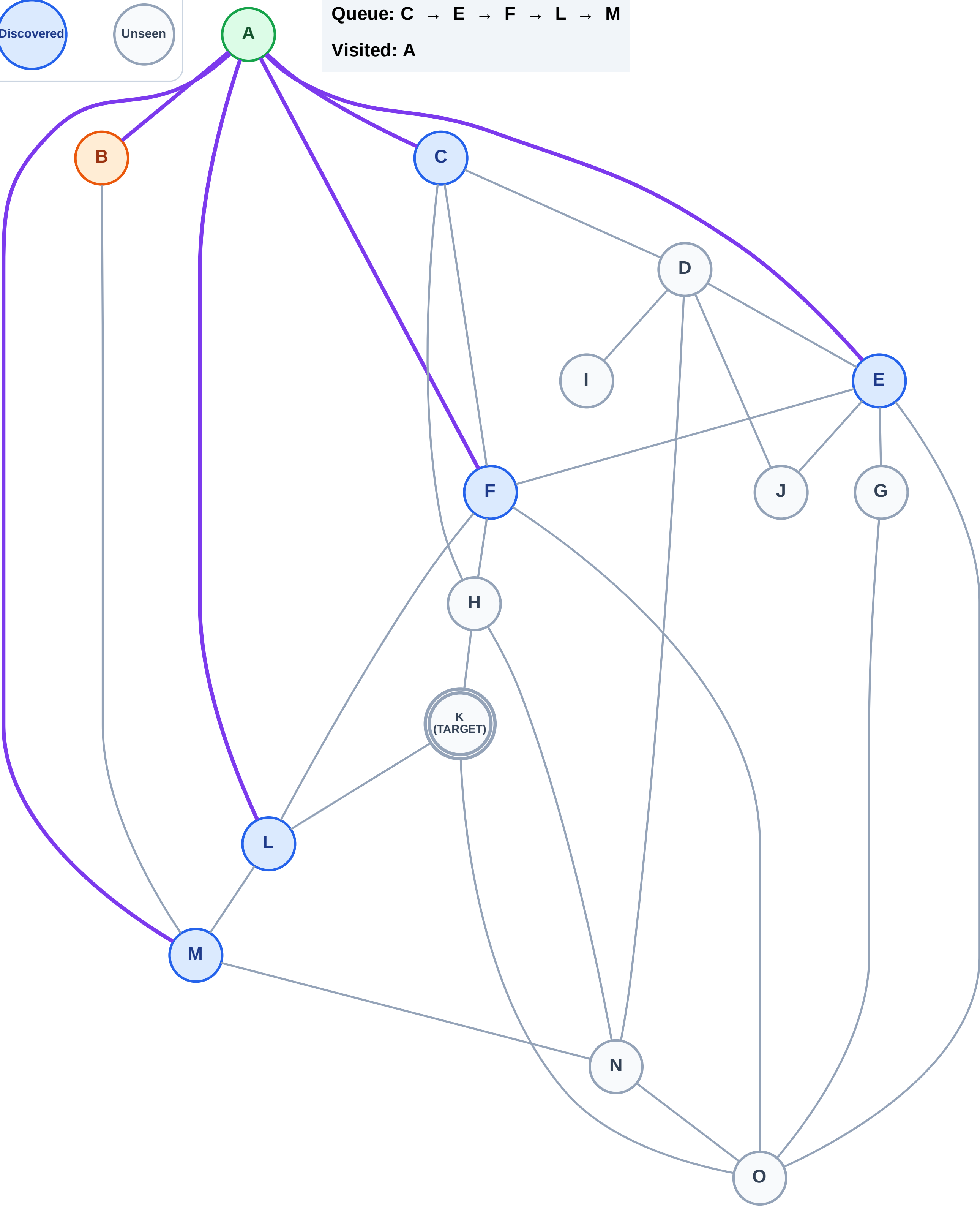
Processing

Discovered

Unseen

Queue: C → E → F → L → M

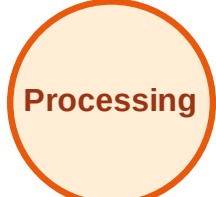
Visited: A



BFS Traversal

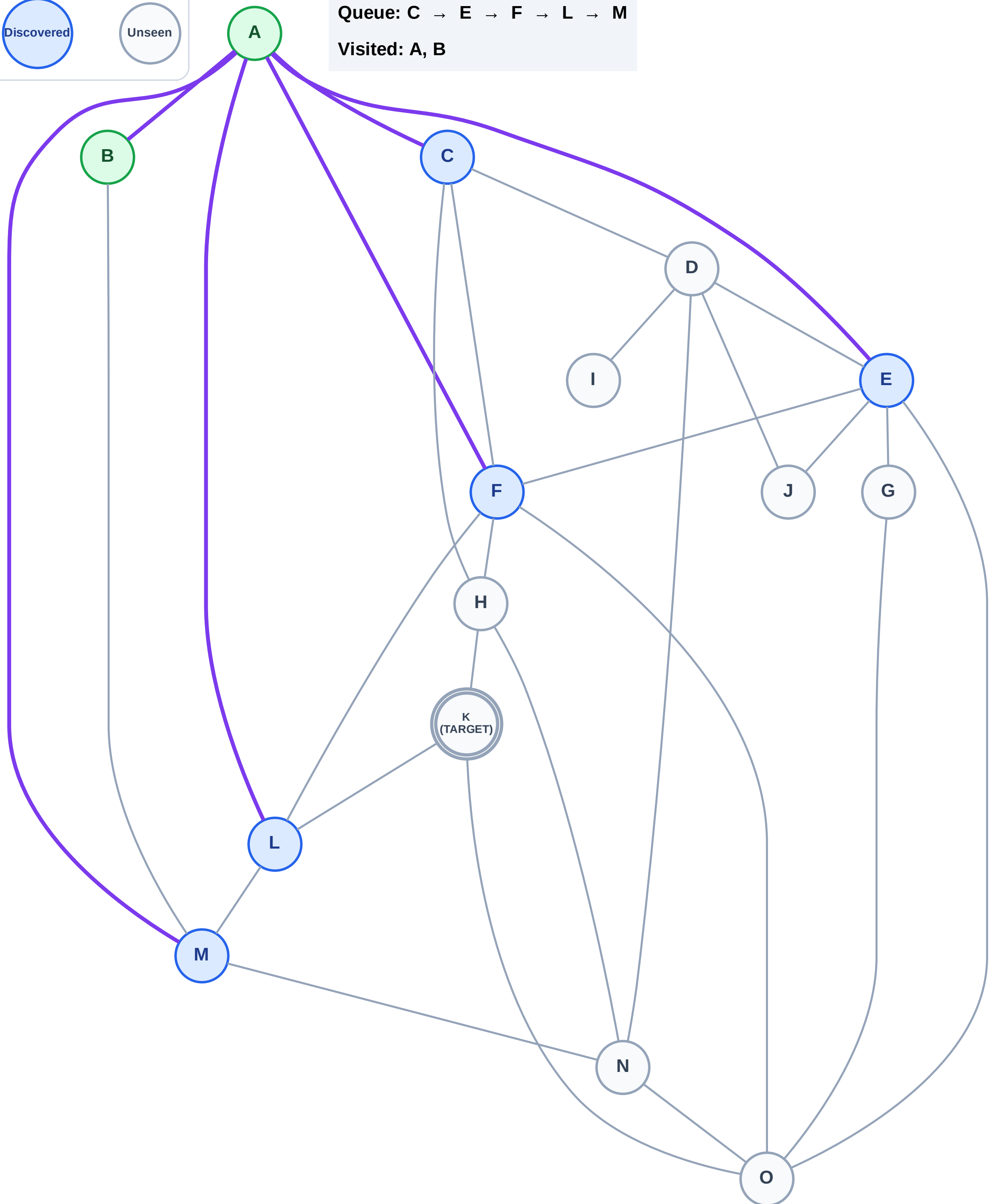
Step 5: No new neighbors from B

Legend



Queue: C → E → F → L → M

Visited: A, B



BFS Traversal

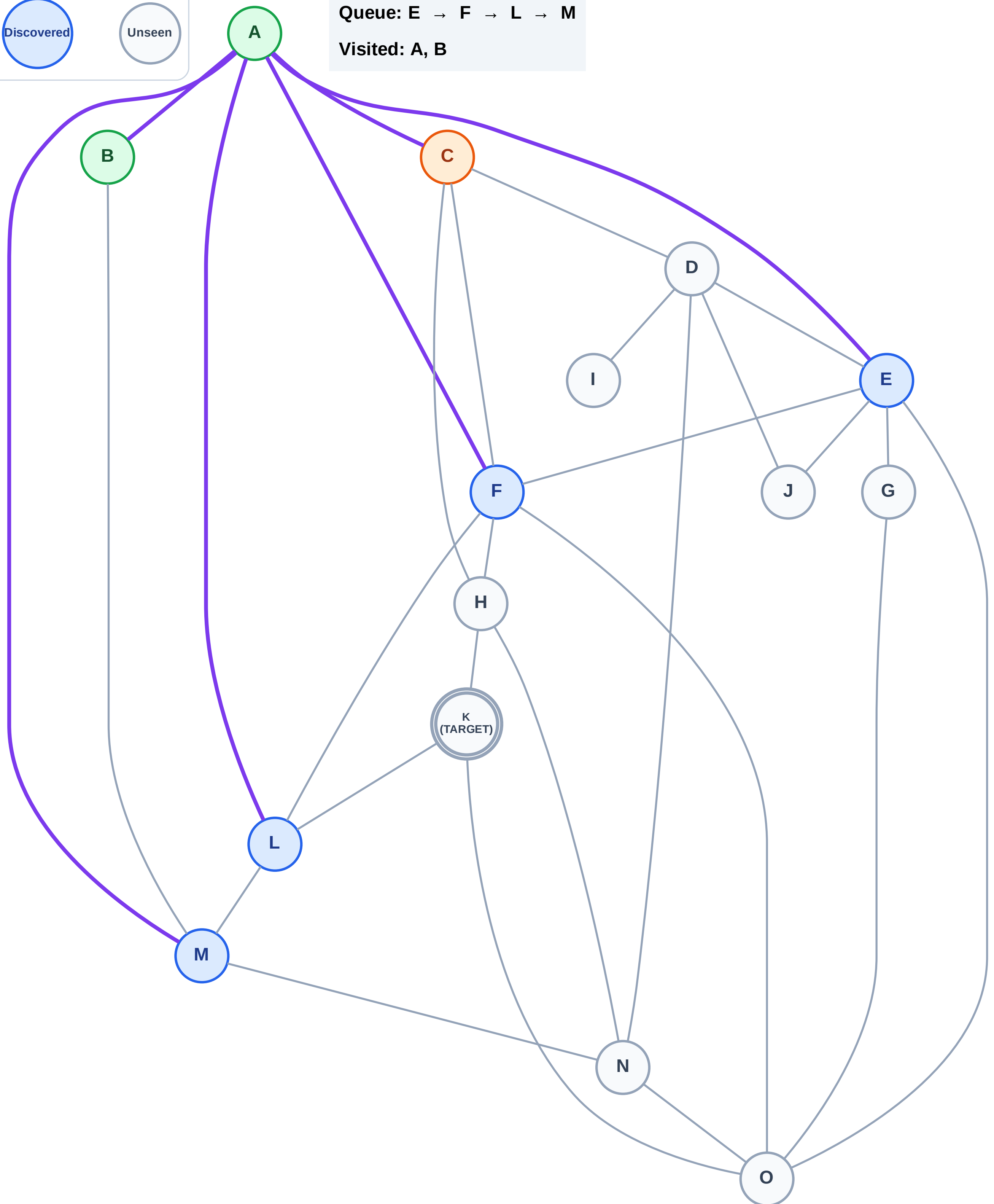
Step 6: Process node C

Legend



Queue: E → F → L → M

Visited: A, B



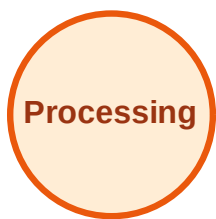
BFS Traversal

Step 7: Discover D, H from C

Legend



Complete



Processing

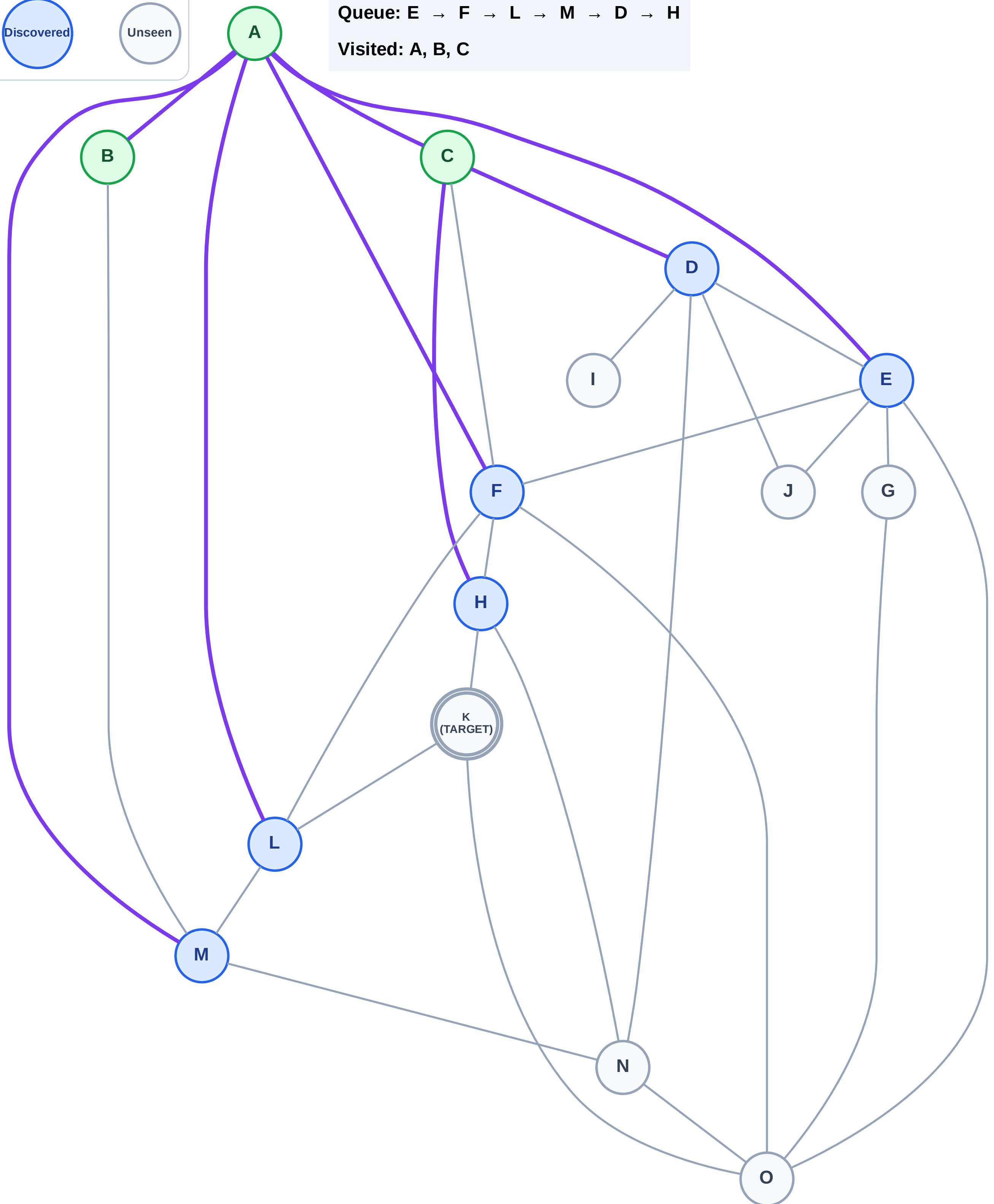


Discovered



Unseen

Queue: E → F → L → M → D → H
Visited: A, B, C



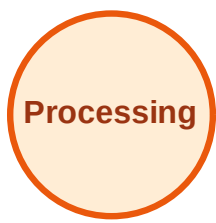
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



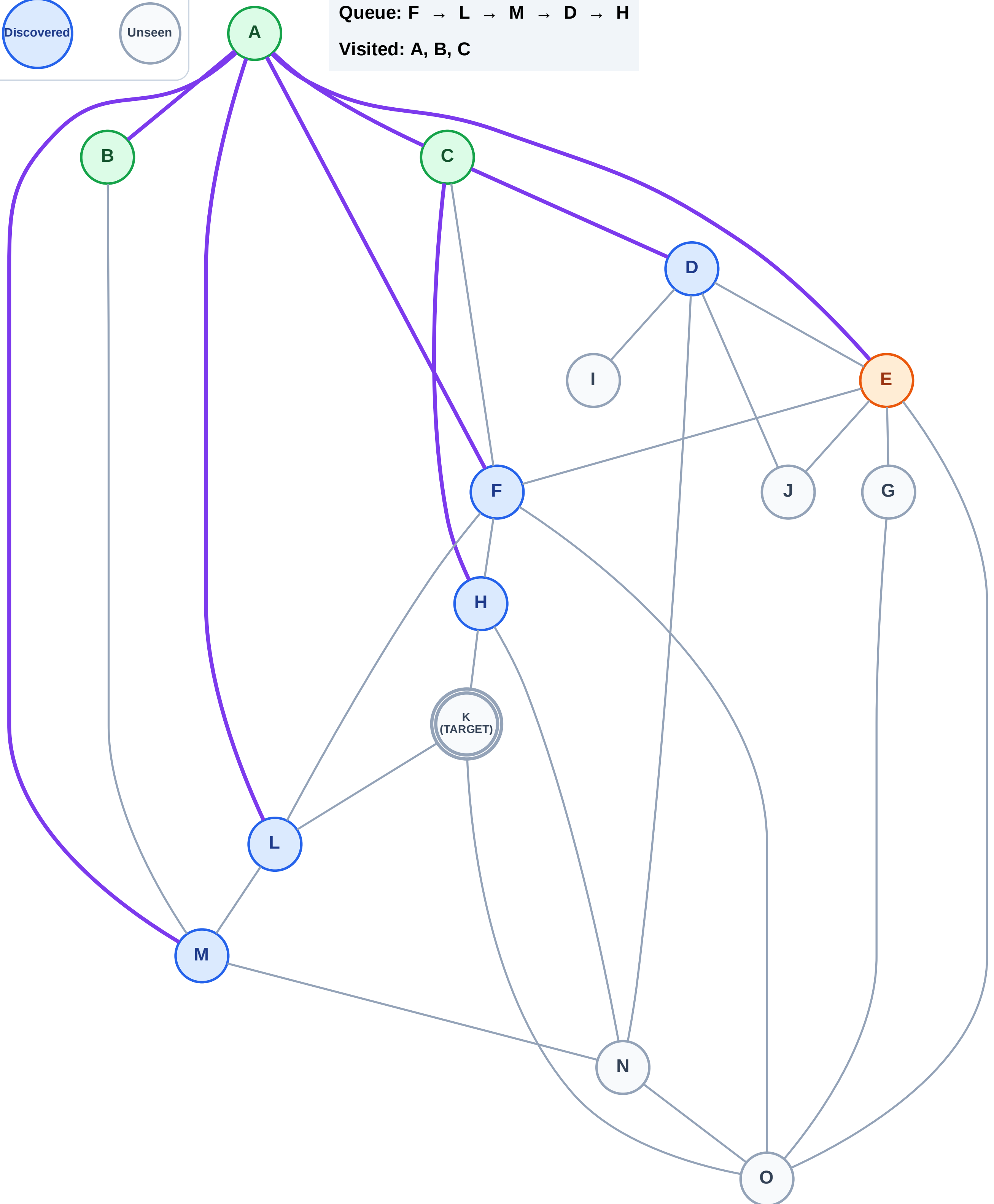
Discovered



Unseen

Queue: F → L → M → D → H

Visited: A, B, C



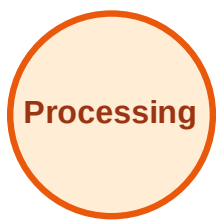
BFS Traversal

Step 9: Discover G, J, O from E

Legend



Complete



Processing



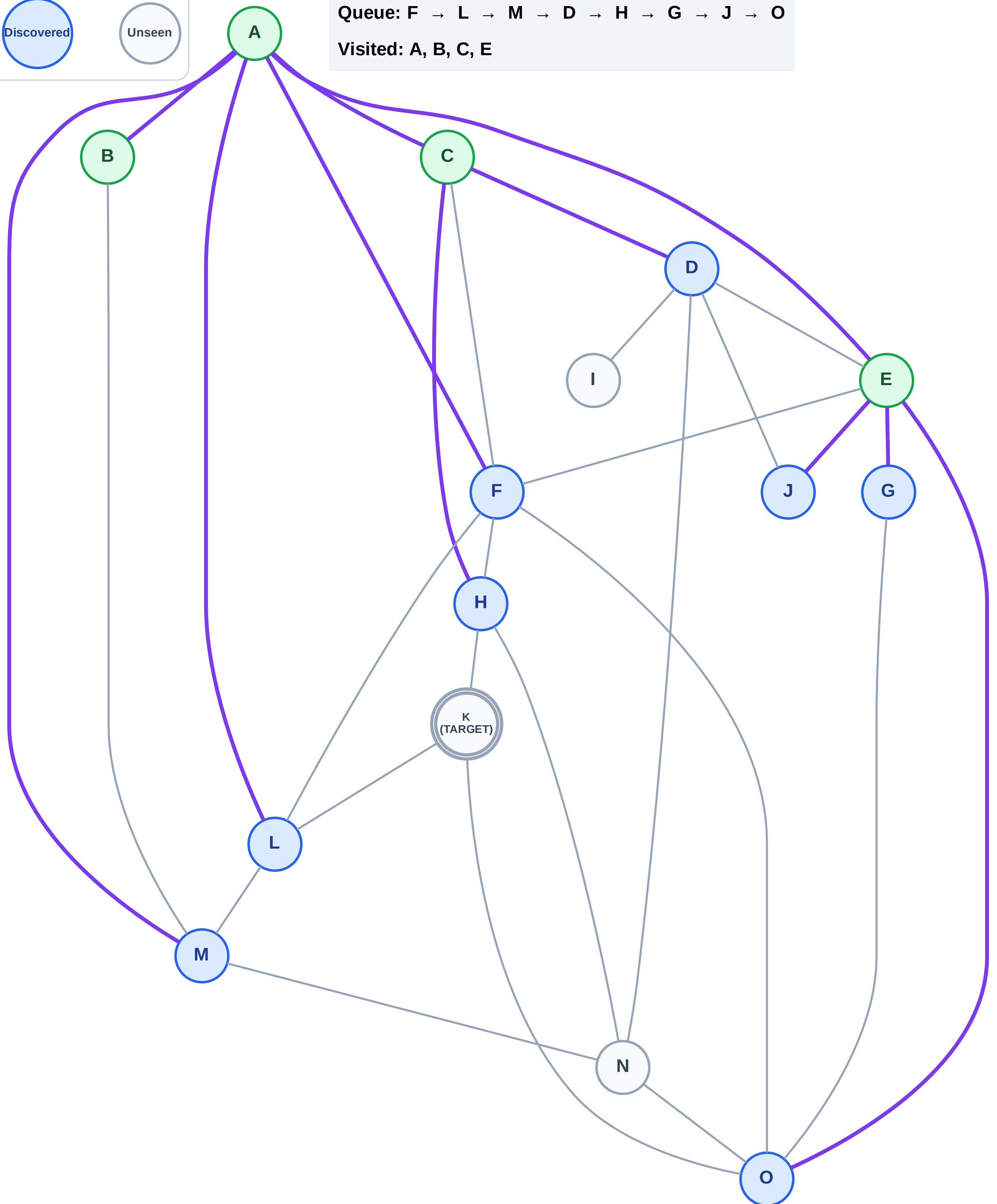
Discovered



Unseen

Queue: F → L → M → D → H → G → J → O

Visited: A, B, C, E



BFS Traversal

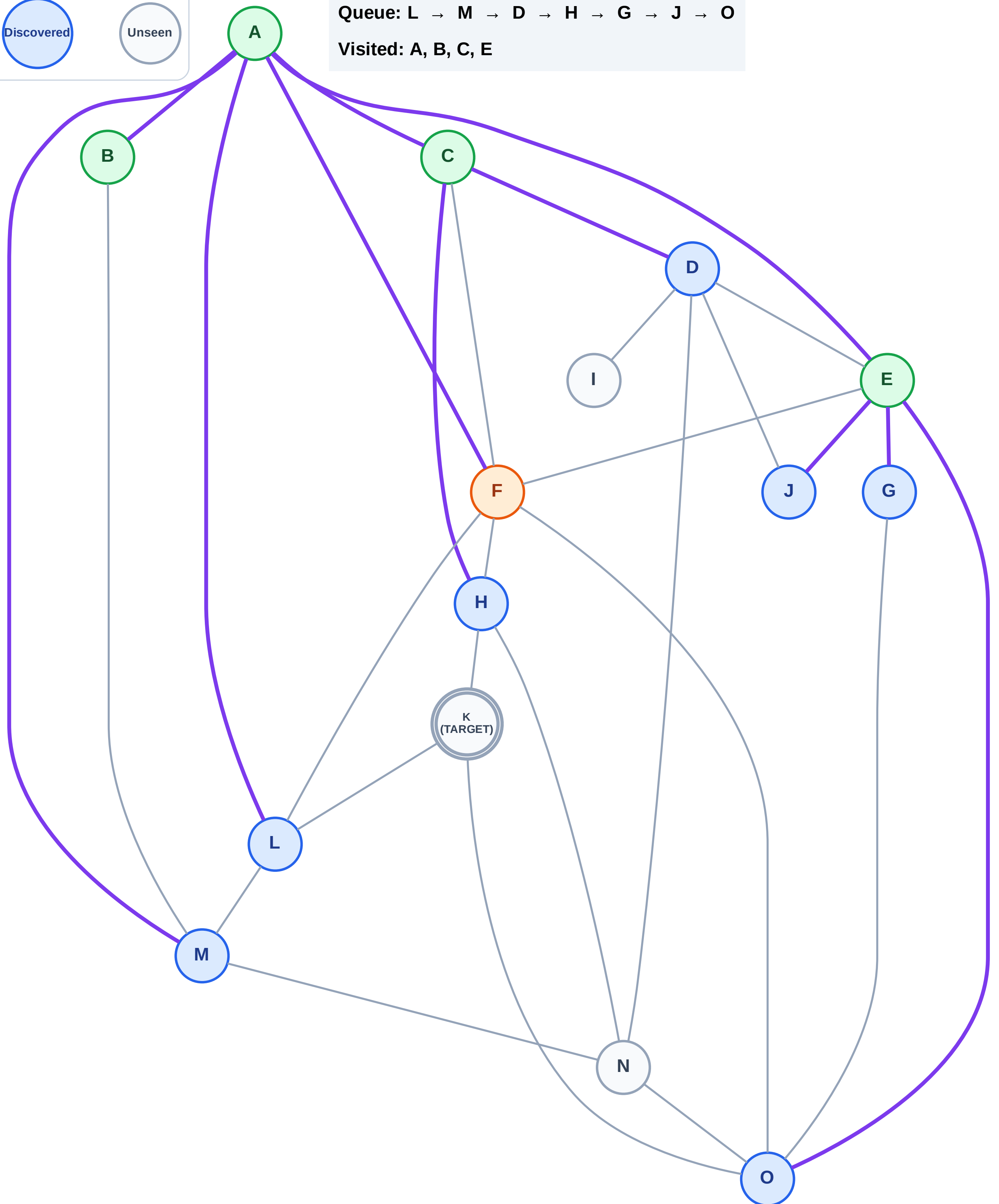
Step 10: Process node F

Legend



Queue: L → M → D → H → G → J → O

Visited: A, B, C, E



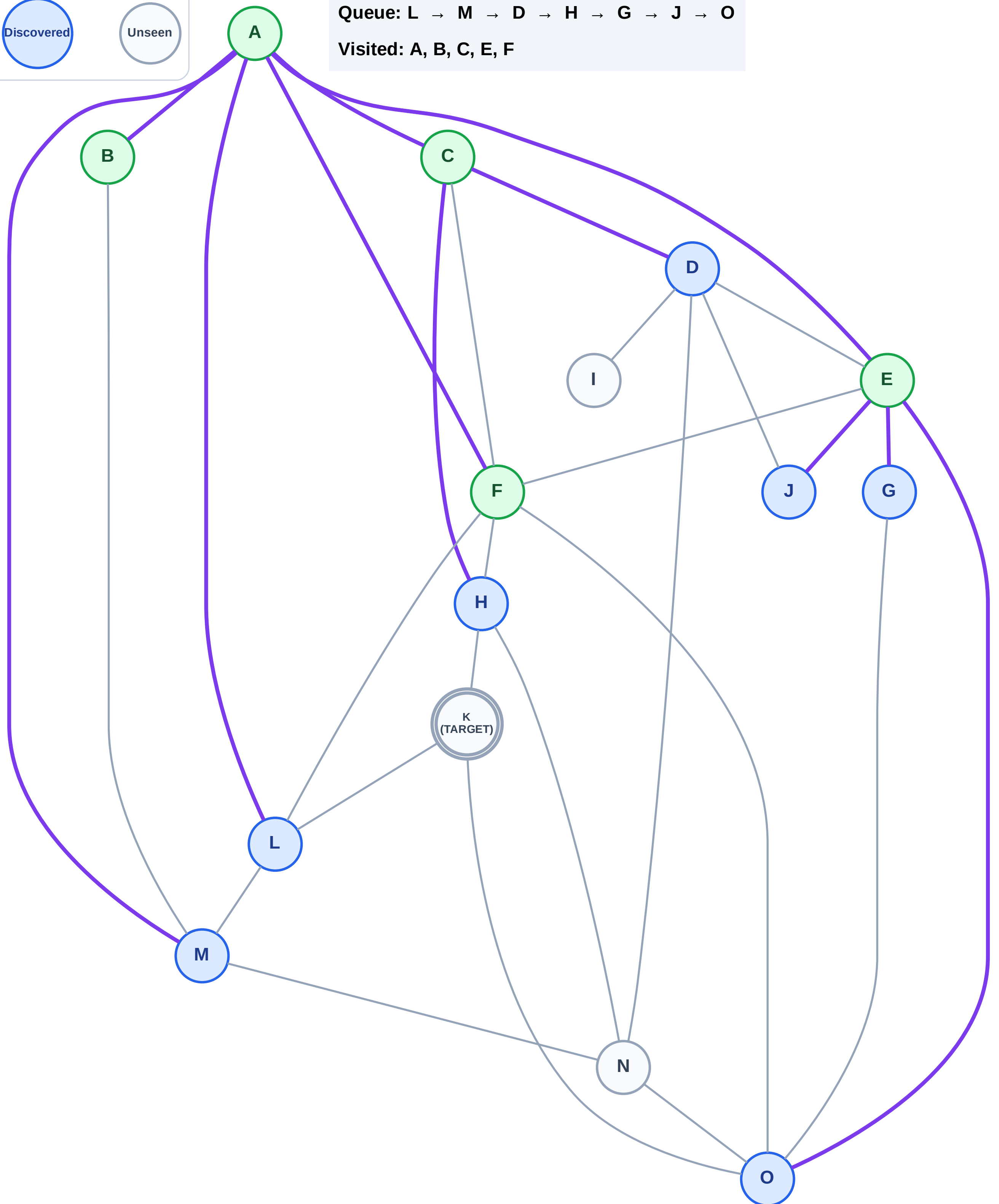
BFS Traversal

Step 11: No new neighbors from F

Legend



Queue: L → M → D → H → G → J → O
Visited: A, B, C, E, F



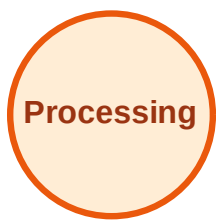
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing



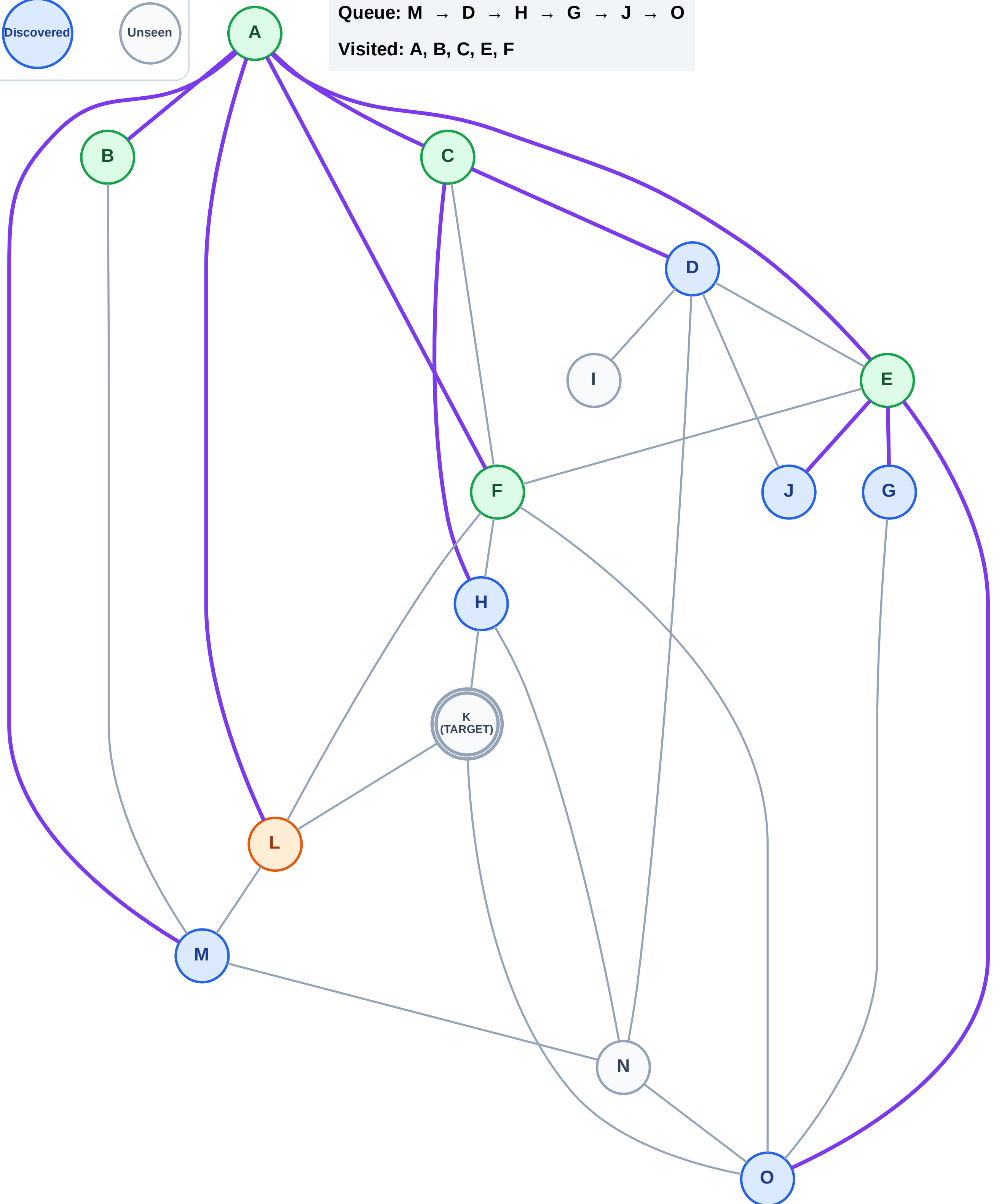
Discovered



Unseen

Queue: M → D → H → G → J → O

Visited: A, B, C, E, F



BFS Traversal

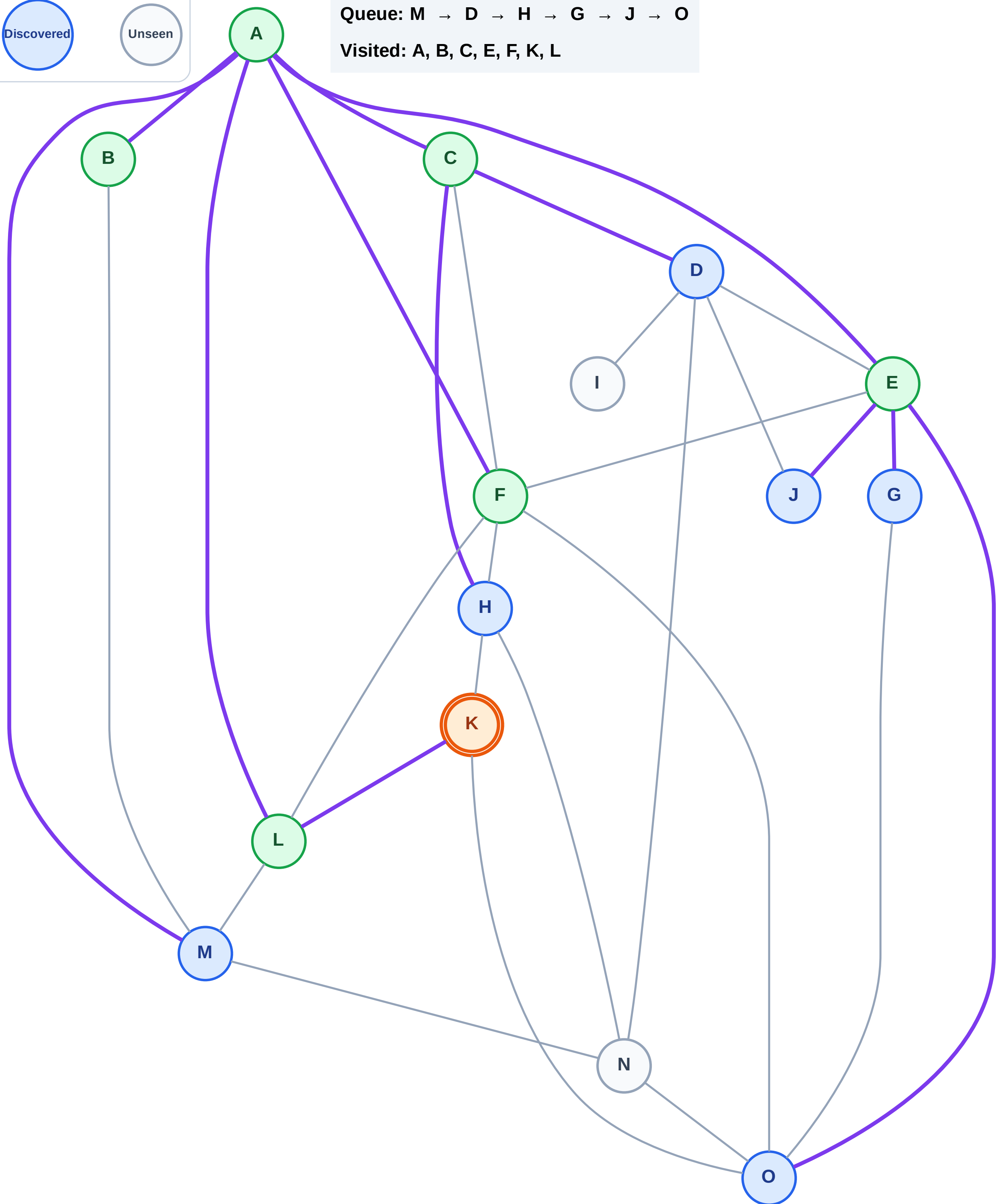
Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → L → K

Legend



Queue: M → D → H → G → J → O

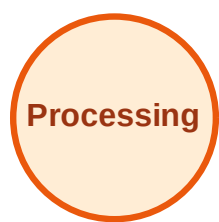
Visited: A, B, C, E, F, K, L



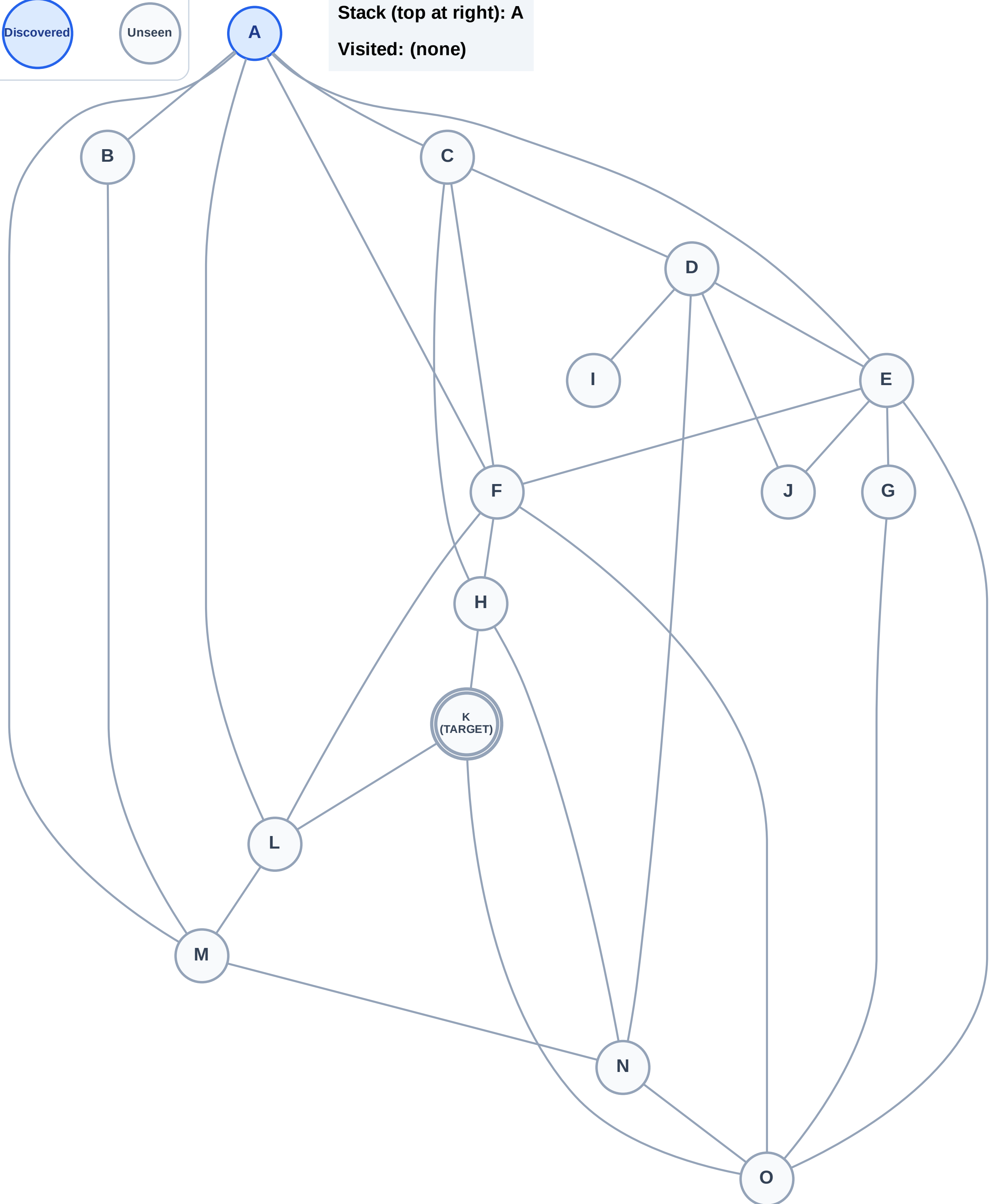
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

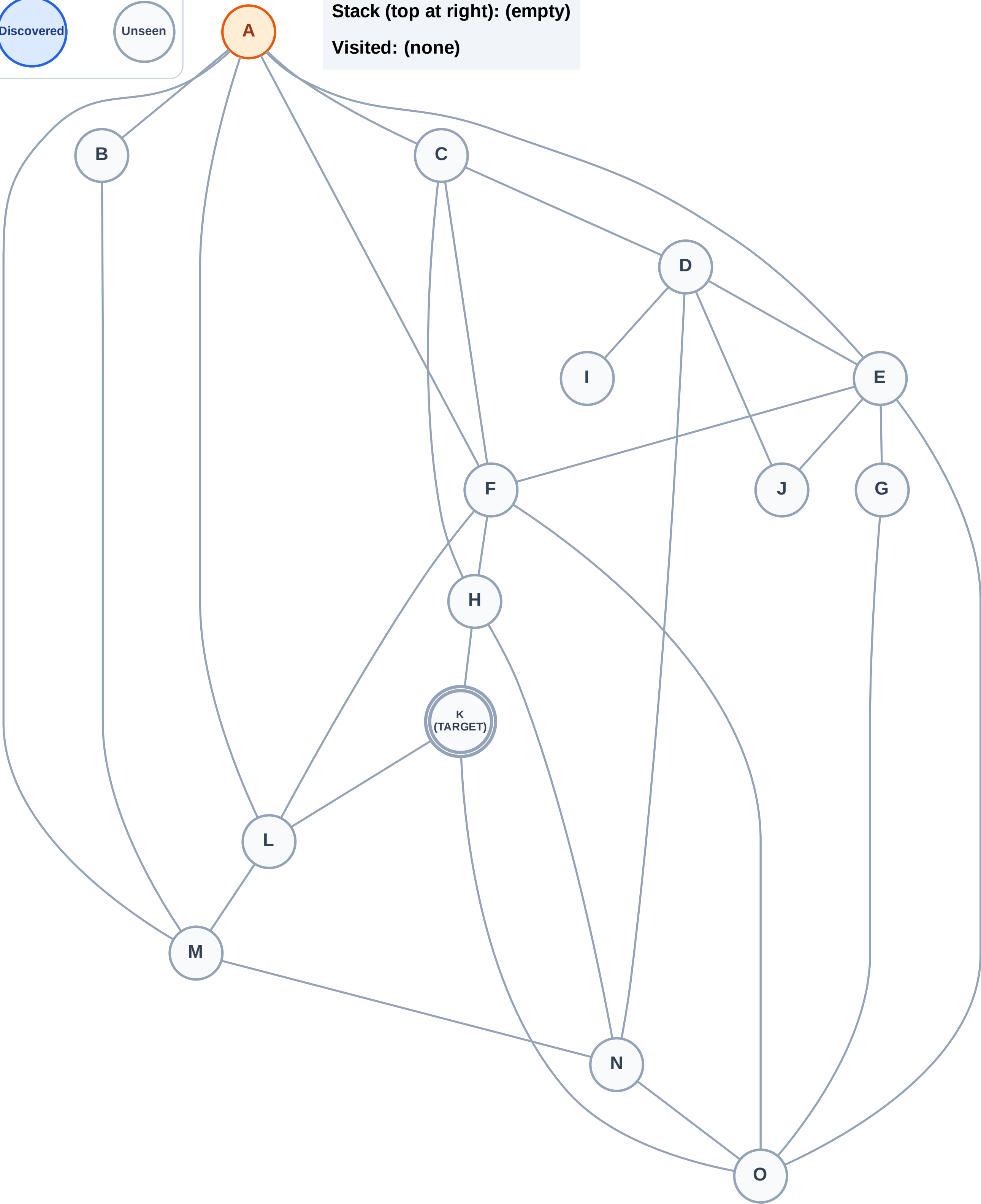
Complete

Processing

Discovered

Unseen

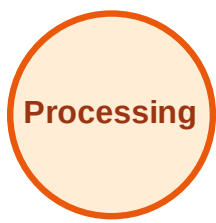
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

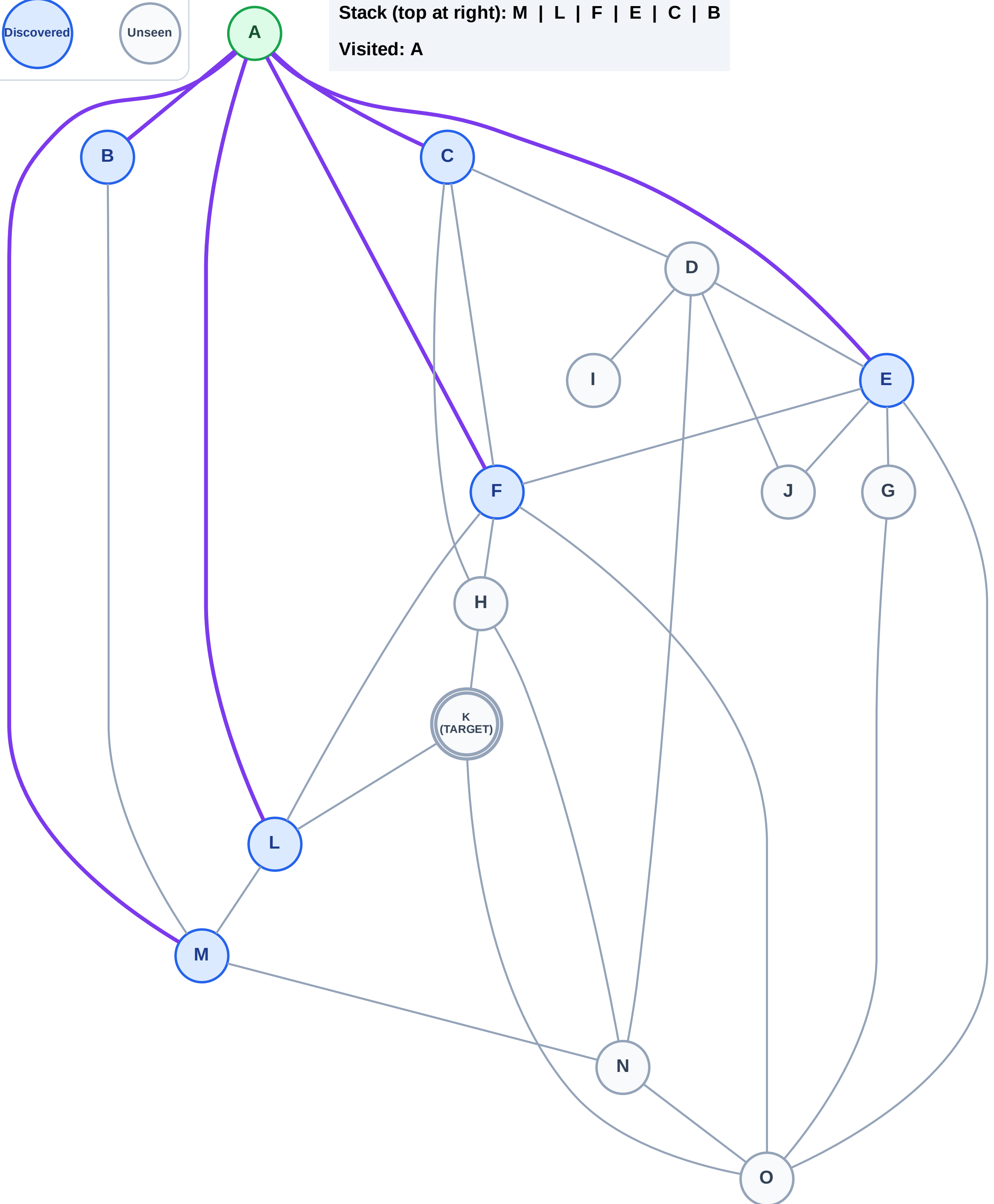
Step 3: Discover M, L, F, E, C, B from A

Legend



Stack (top at right): M | L | F | E | C | B

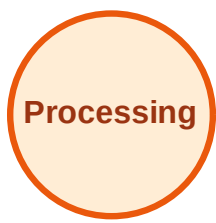
Visited: A



DFS Traversal

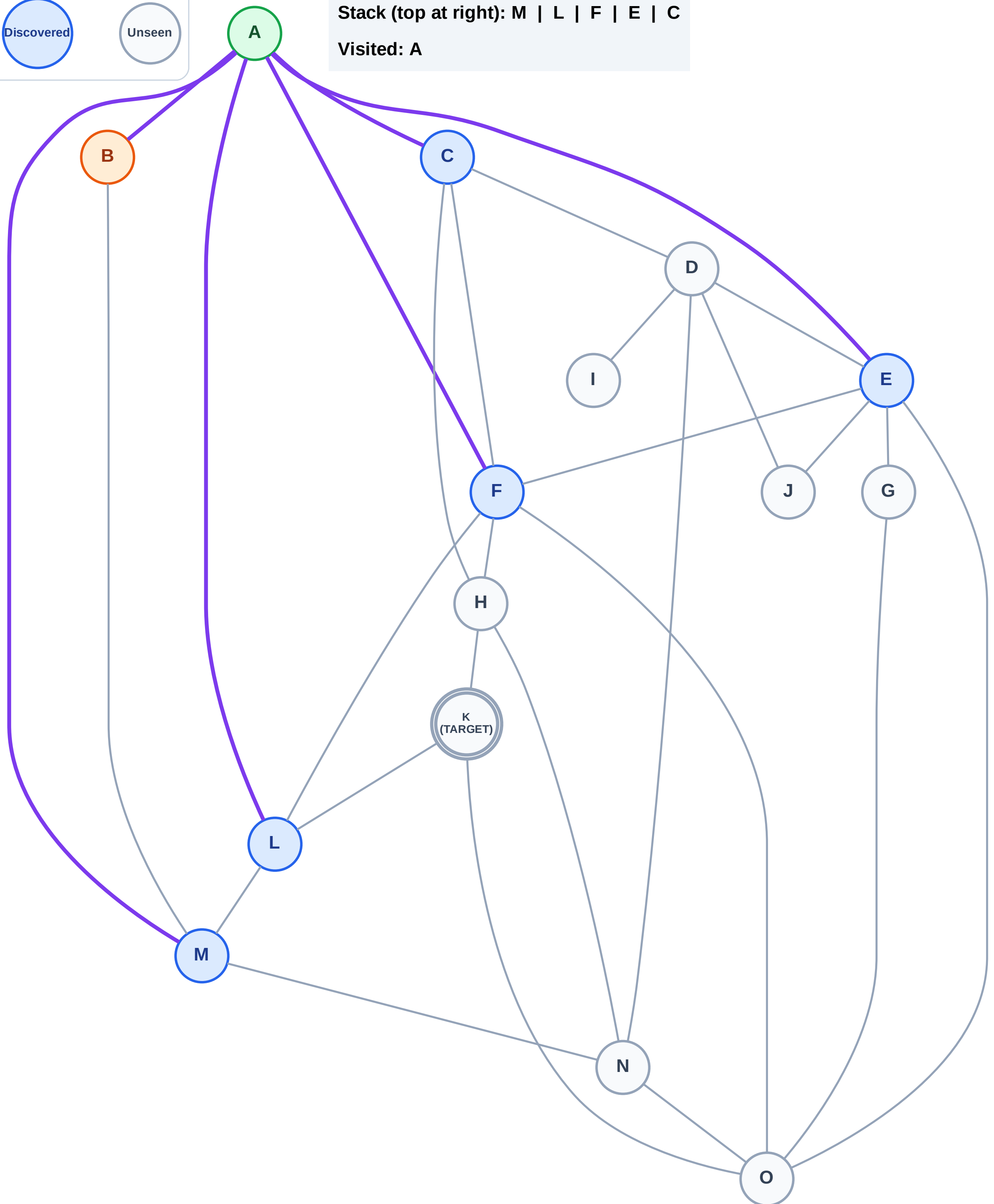
Step 4: Process node B

Legend



Stack (top at right): M | L | F | E | C

Visited: A

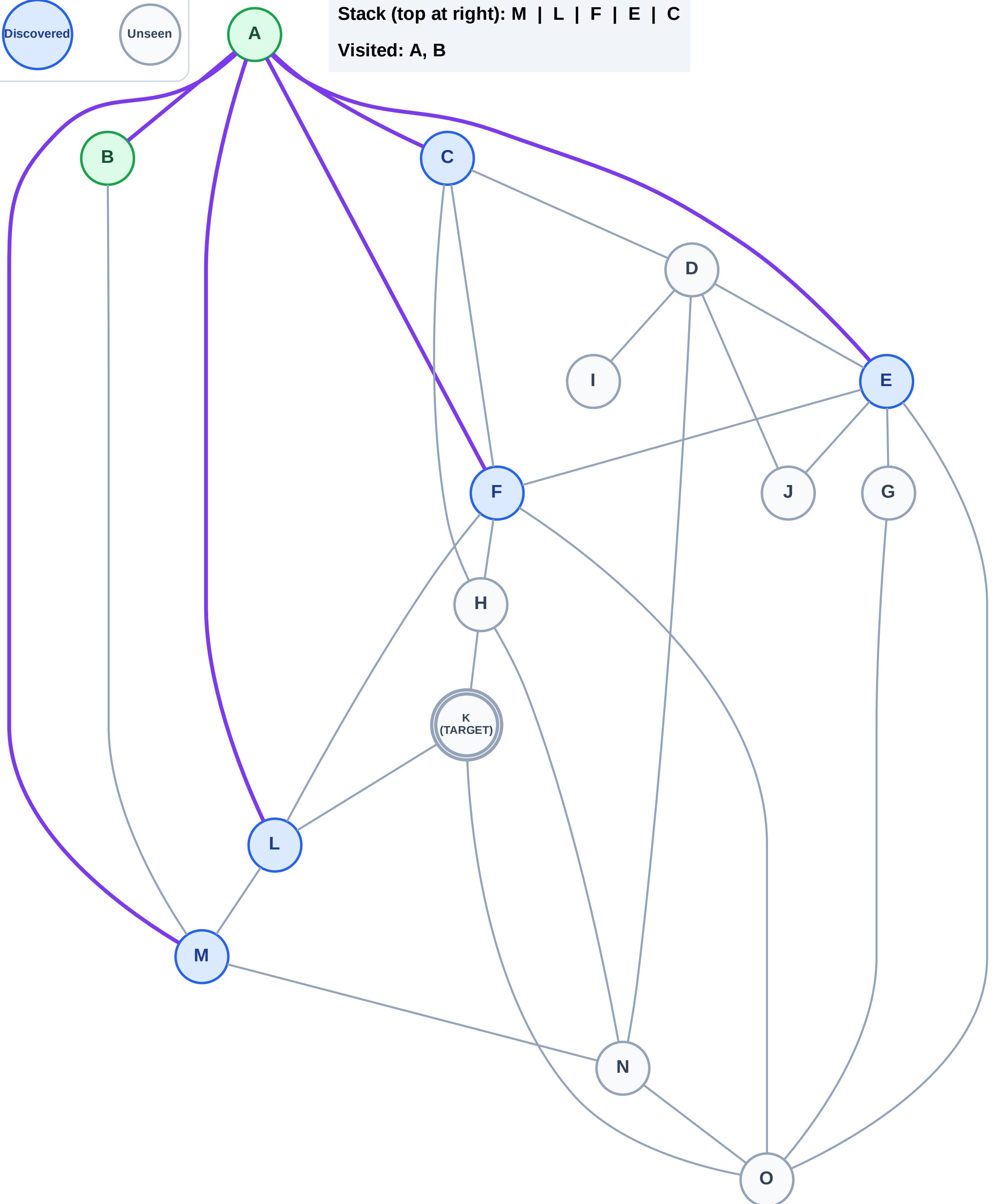


Step 5: No new neighbors from B

Stack (top at right): M | L | F | E | C

Legend

Unseen



DFS Traversal

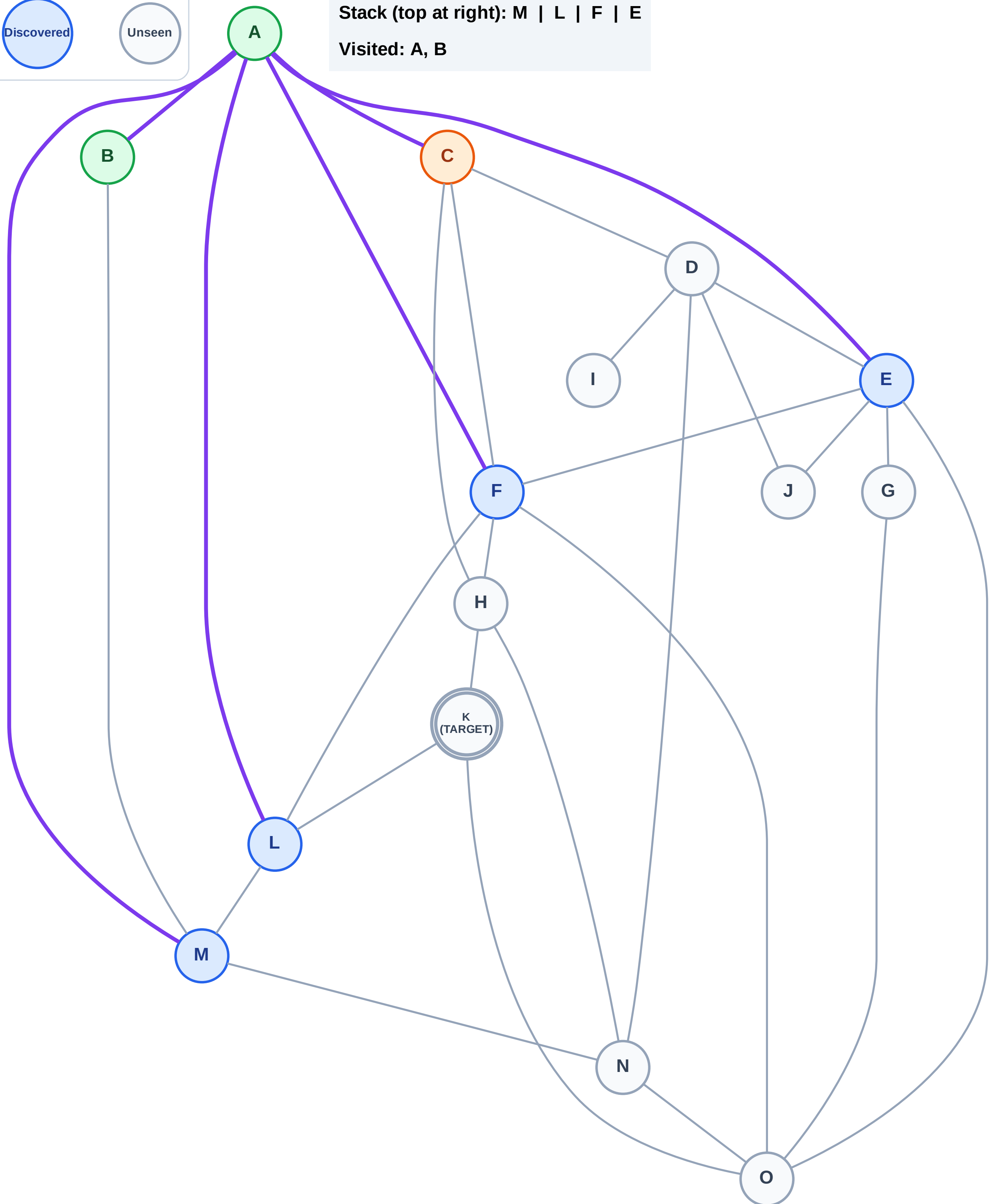
Step 6: Process node C

Legend



Stack (top at right): M | L | F | E

Visited: A, B

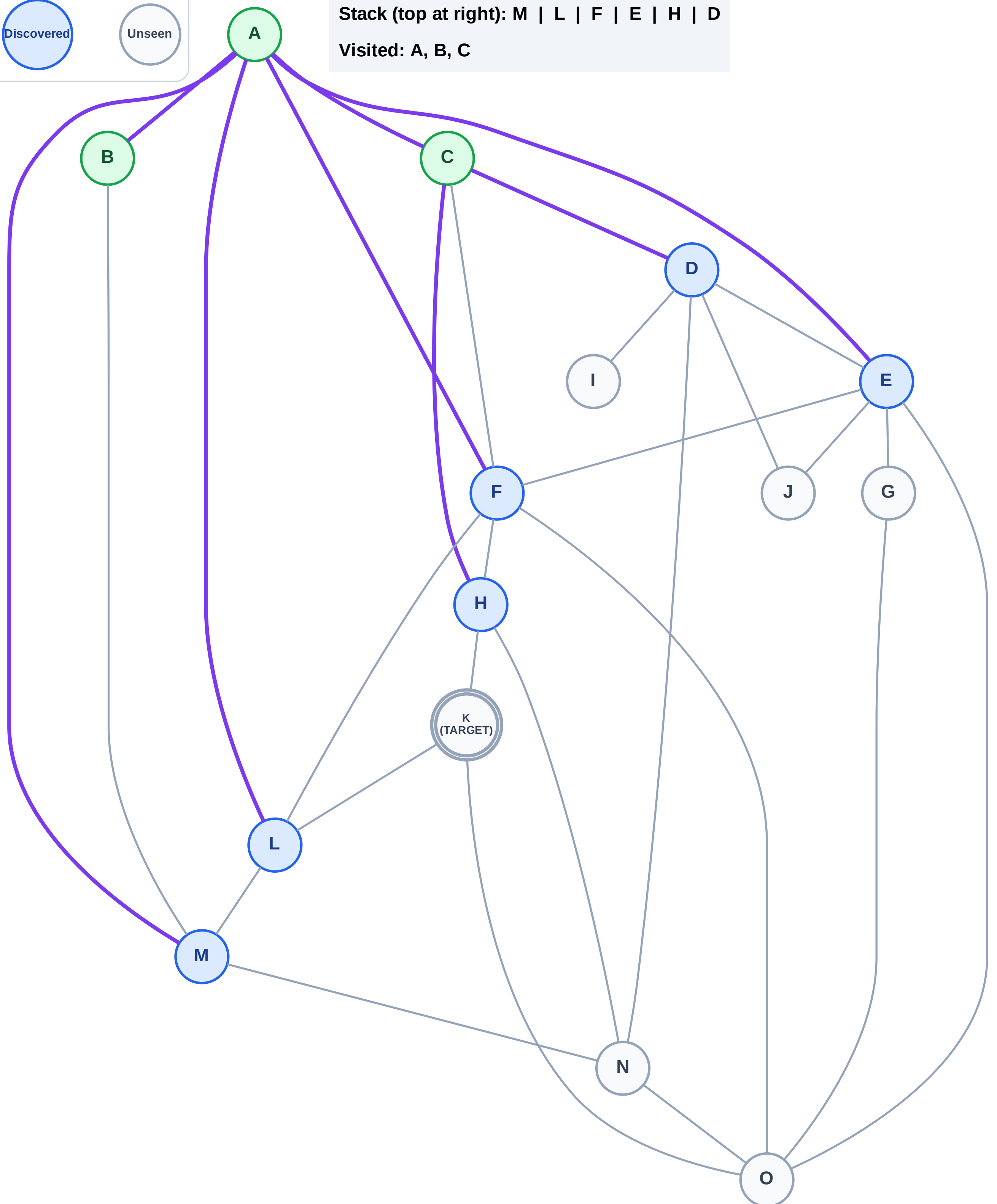


Step 7: Discover H, D from C

Stack (top at right): M | L | F | E | H | D

Legend

Unseen



Step 8: Process node D

Legend

Complete

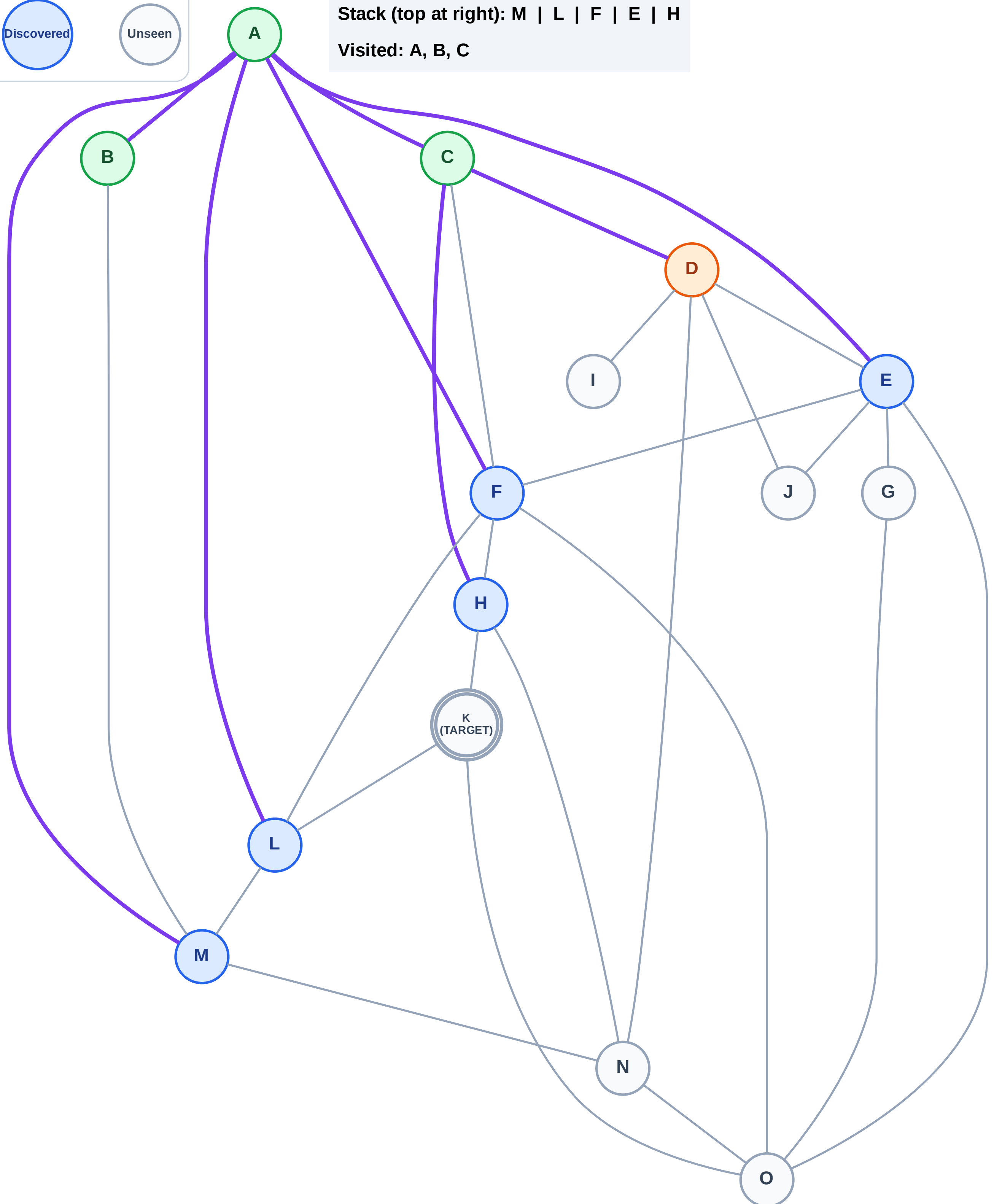
Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H

Visited: A, B, C



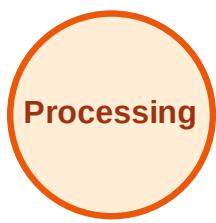
DFS Traversal

Step 9: Discover N, J, I from D

Legend



Complete



Processing



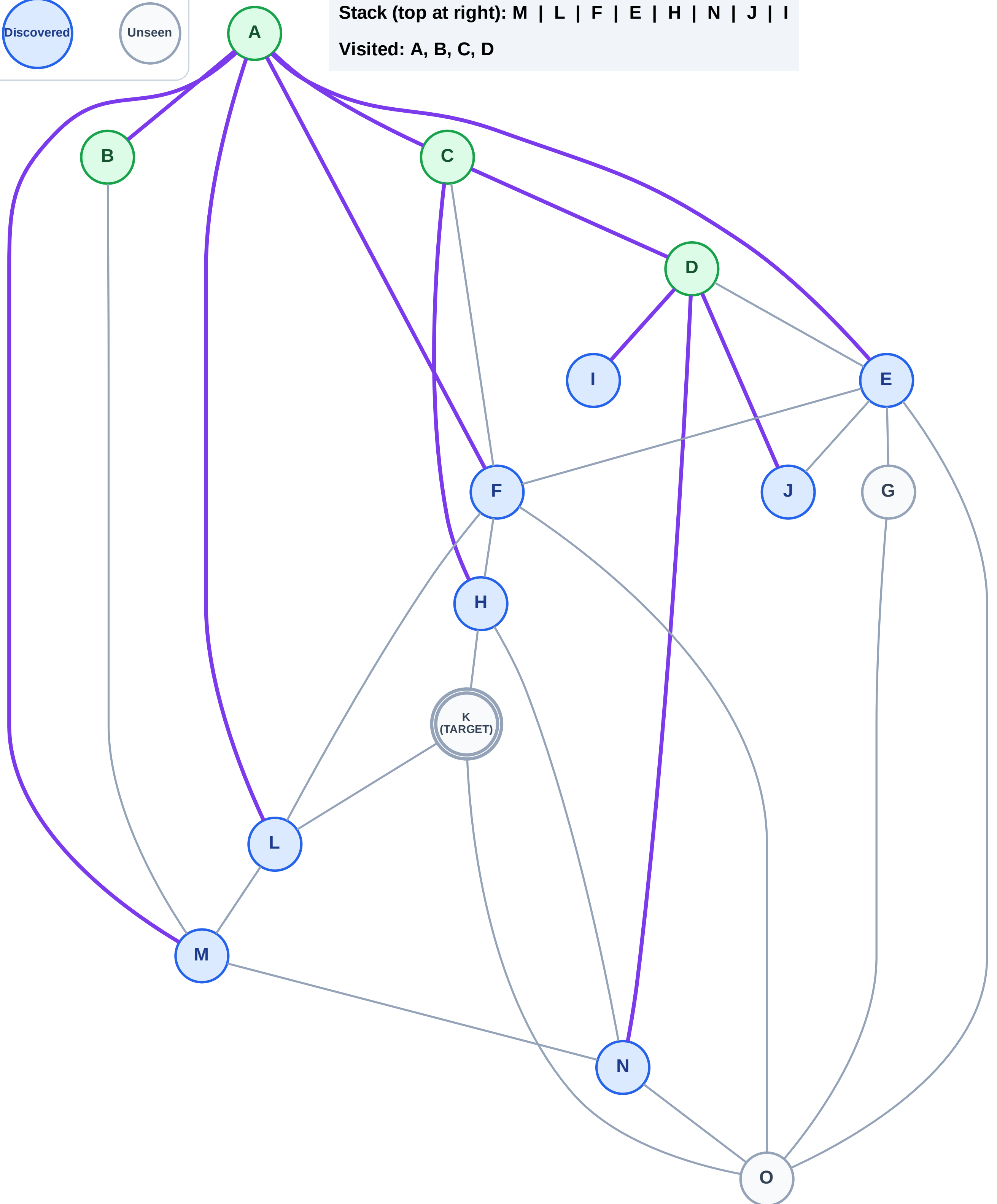
Discovered



Unseen

Stack (top at right): M | L | F | E | H | N | J | I

Visited: A, B, C, D



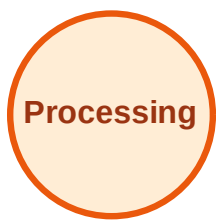
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



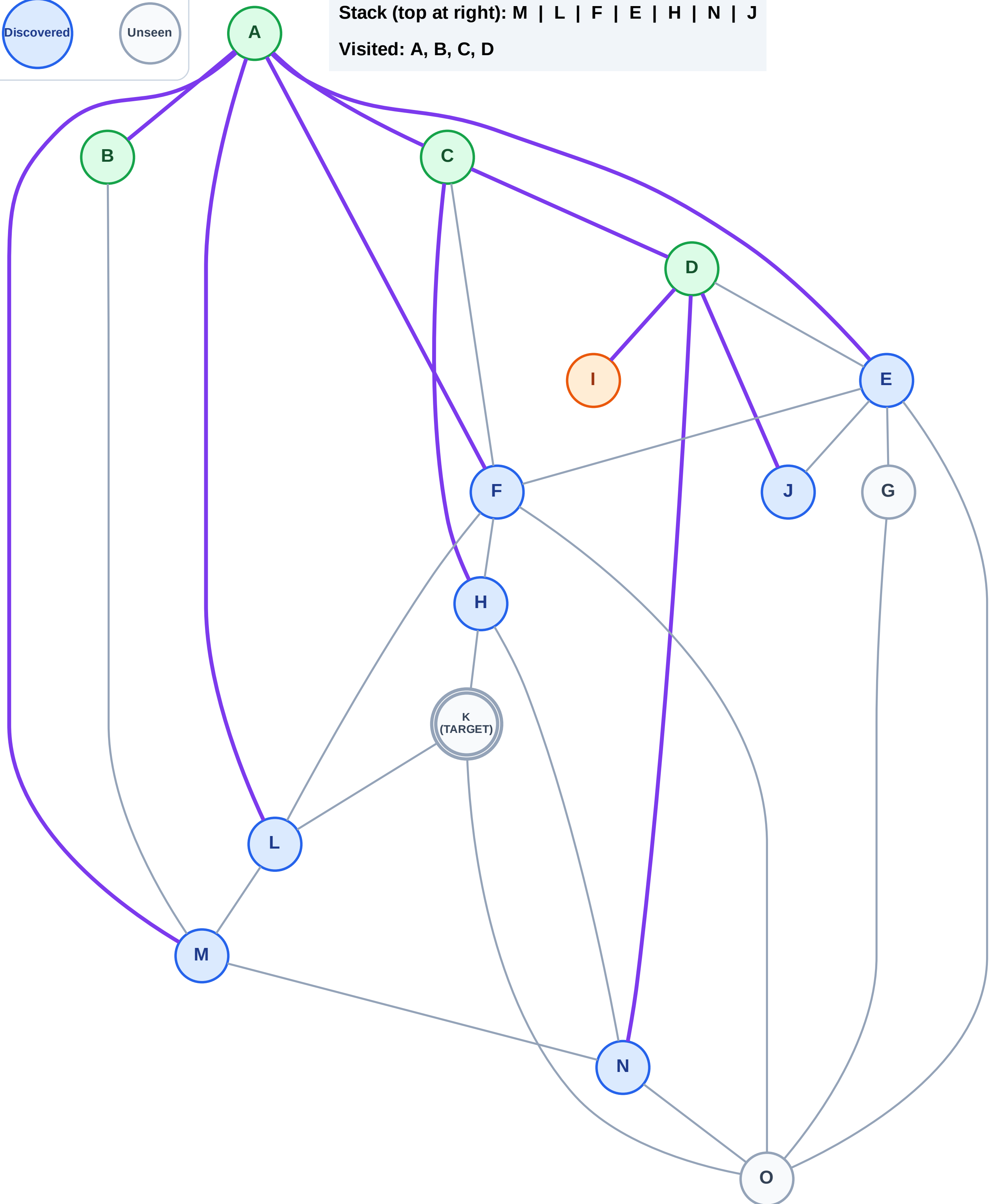
Discovered



Unseen

Stack (top at right): M | L | F | E | H | N | J

Visited: A, B, C, D



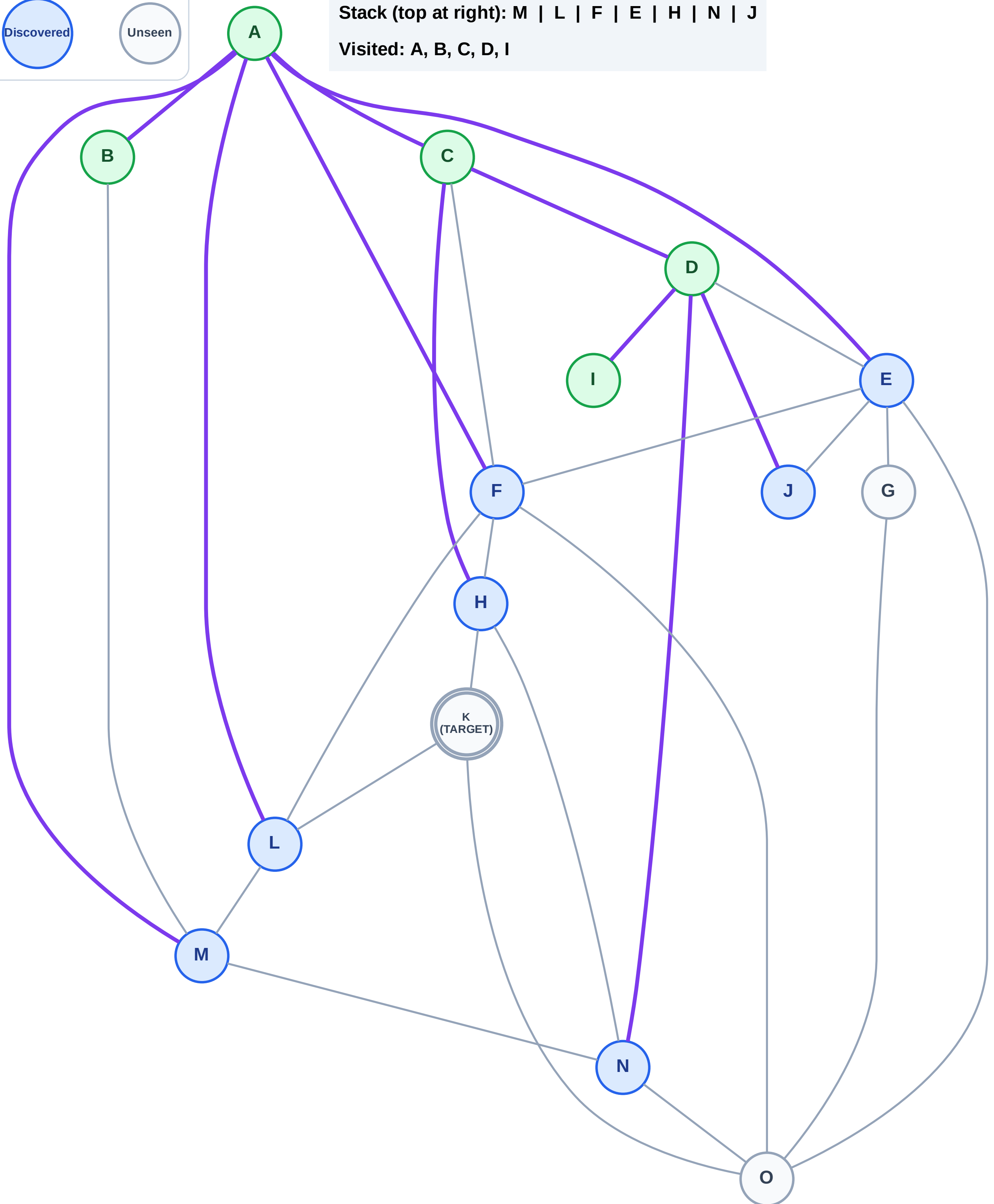
DFS Traversal

Step 11: No new neighbors from I

Legend



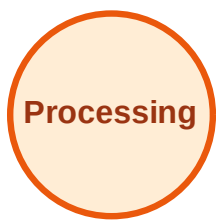
Stack (top at right): M | L | F | E | H | N | J
Visited: A, B, C, D, I



DFS Traversal

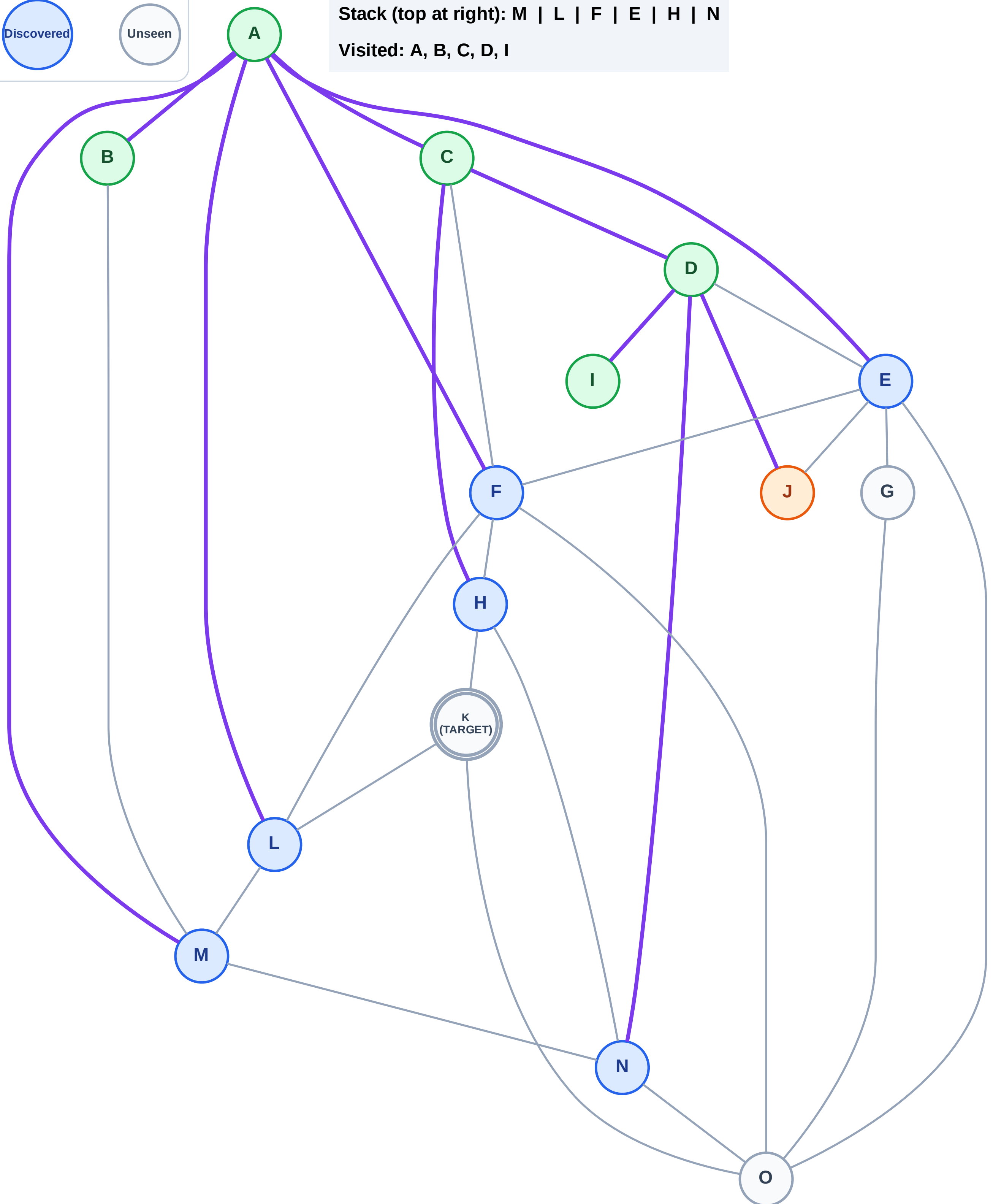
Step 12: Process node J

Legend



Stack (top at right): M | L | F | E | H | N

Visited: A, B, C, D, I



DFS Traversal

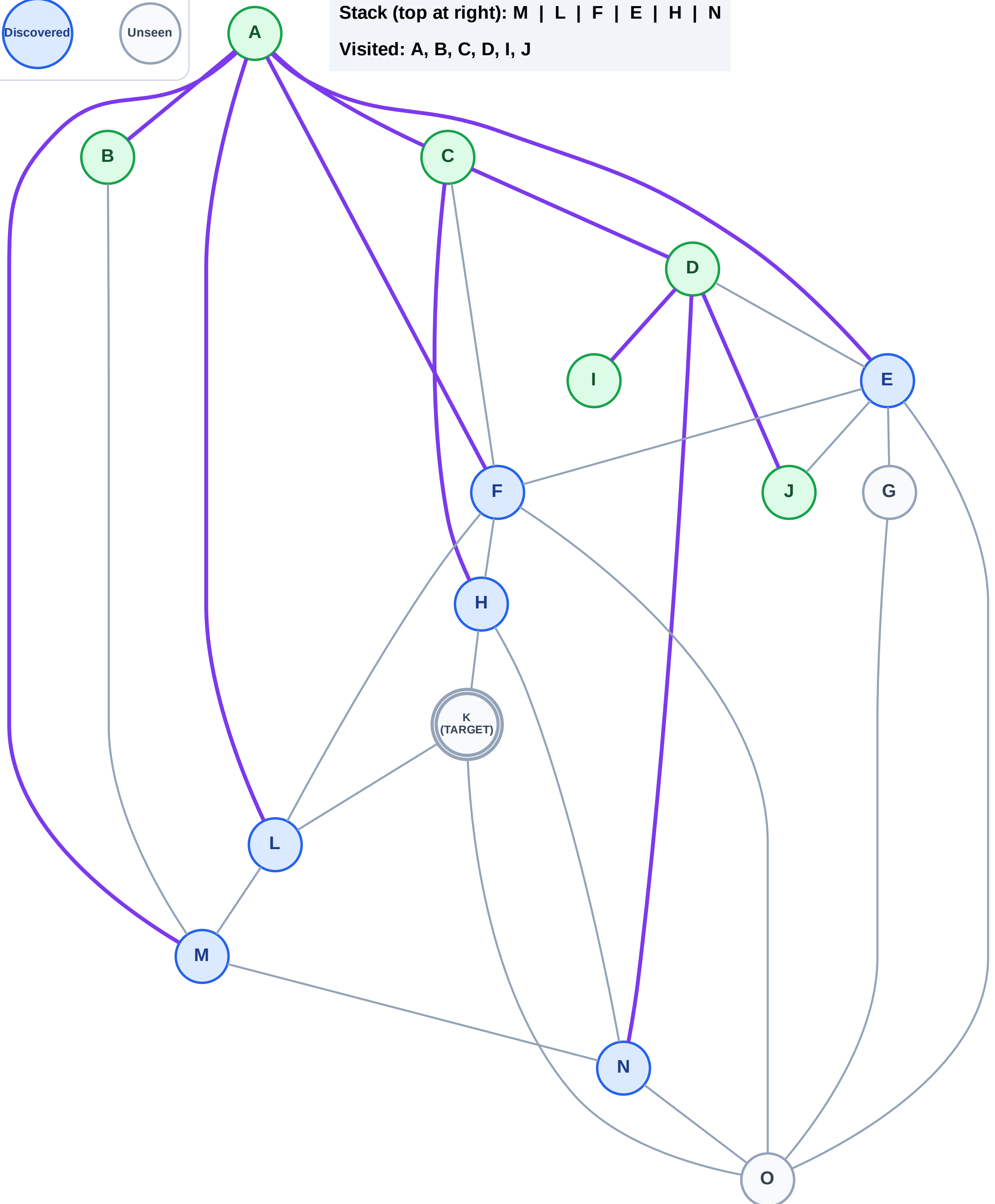
Step 13: No new neighbors from J

Legend



Stack (top at right): M | L | F | E | H | N

Visited: A, B, C, D, I, J



DFS Traversal

Step 14: Process node N

Legend

Complete

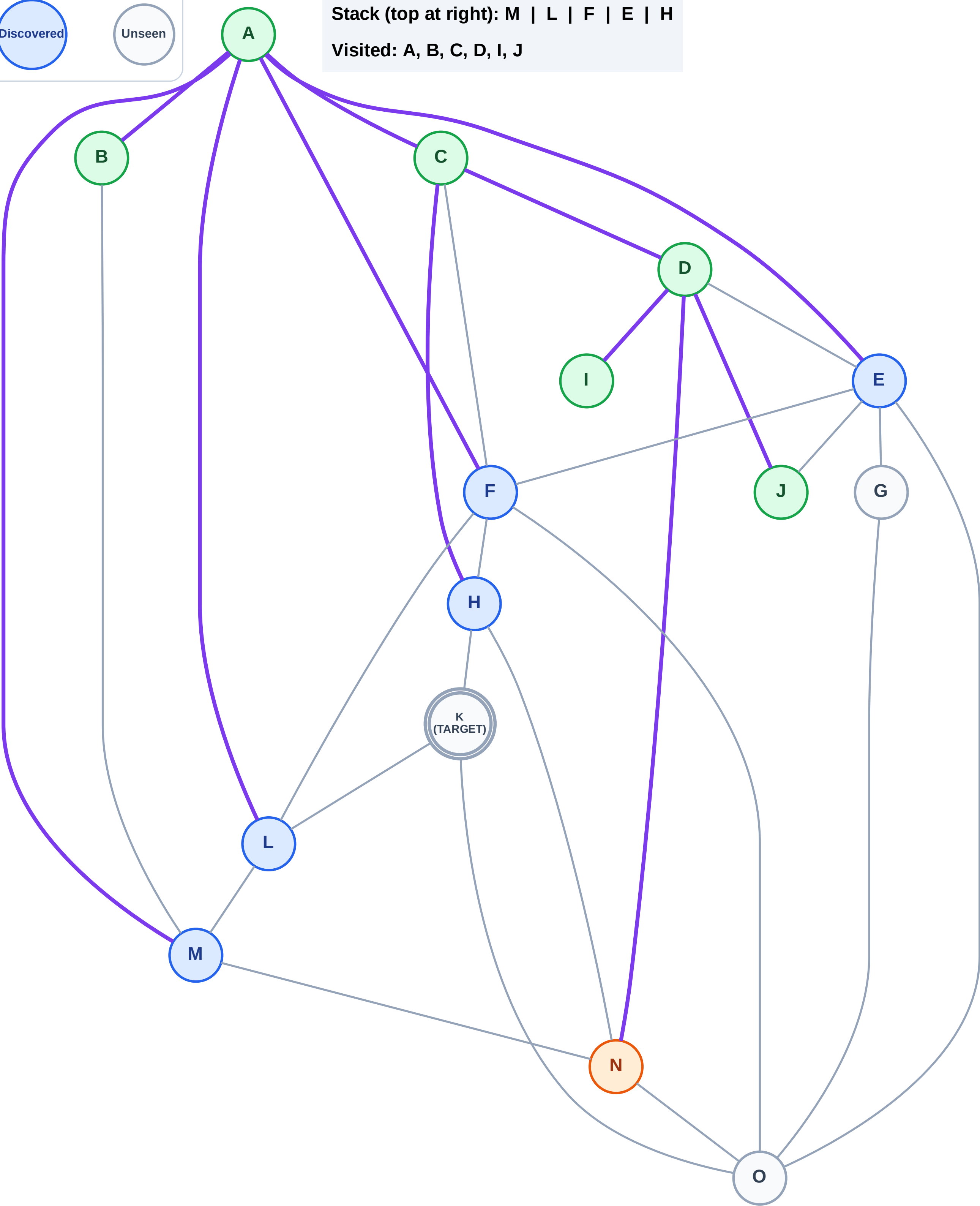
Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H

Visited: A, B, C, D, I, J



DFS Traversal

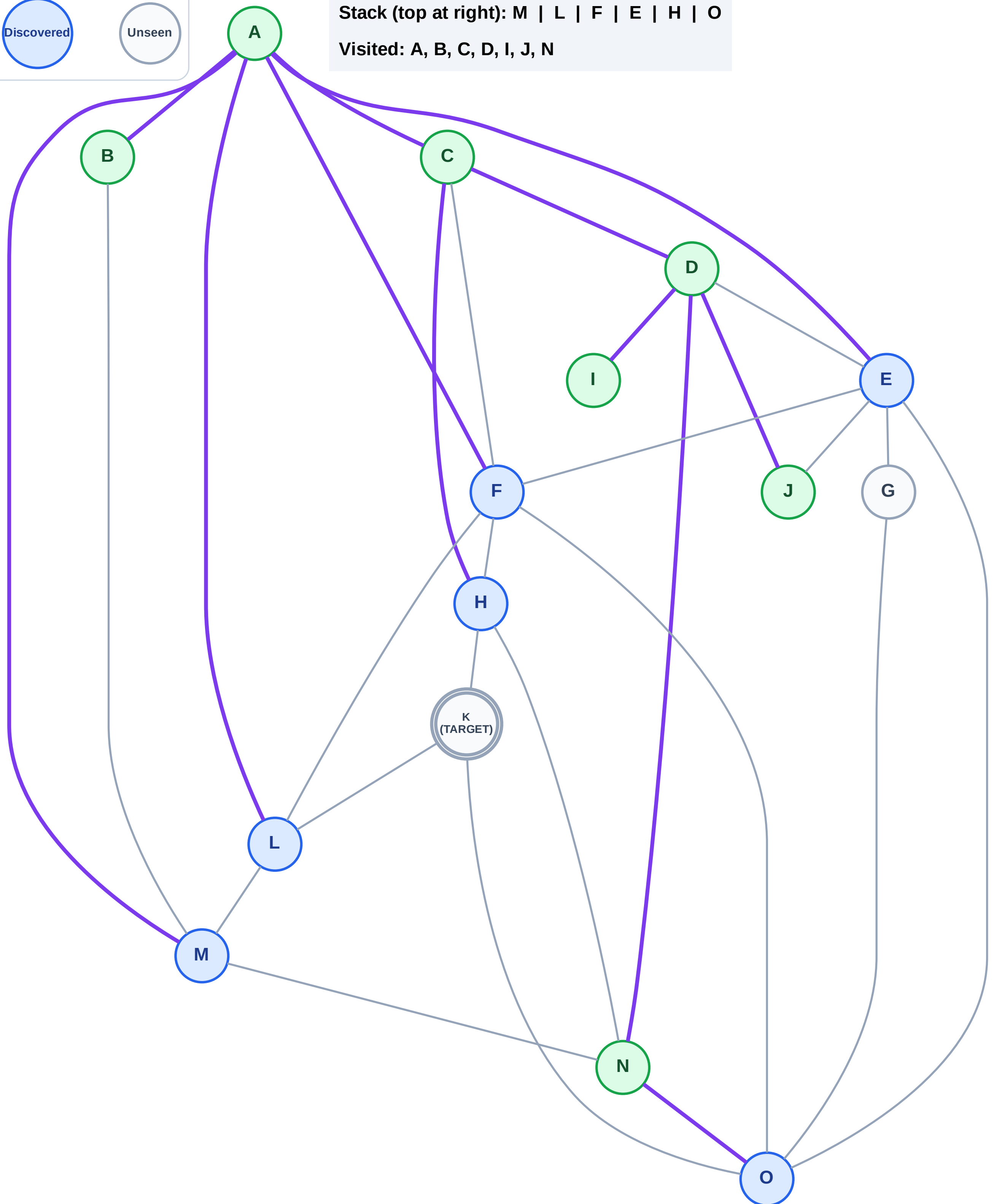
Step 15: Discover O from N

Legend



Stack (top at right): M | L | F | E | H | O

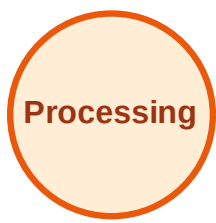
Visited: A, B, C, D, I, J, N



DFS Traversal

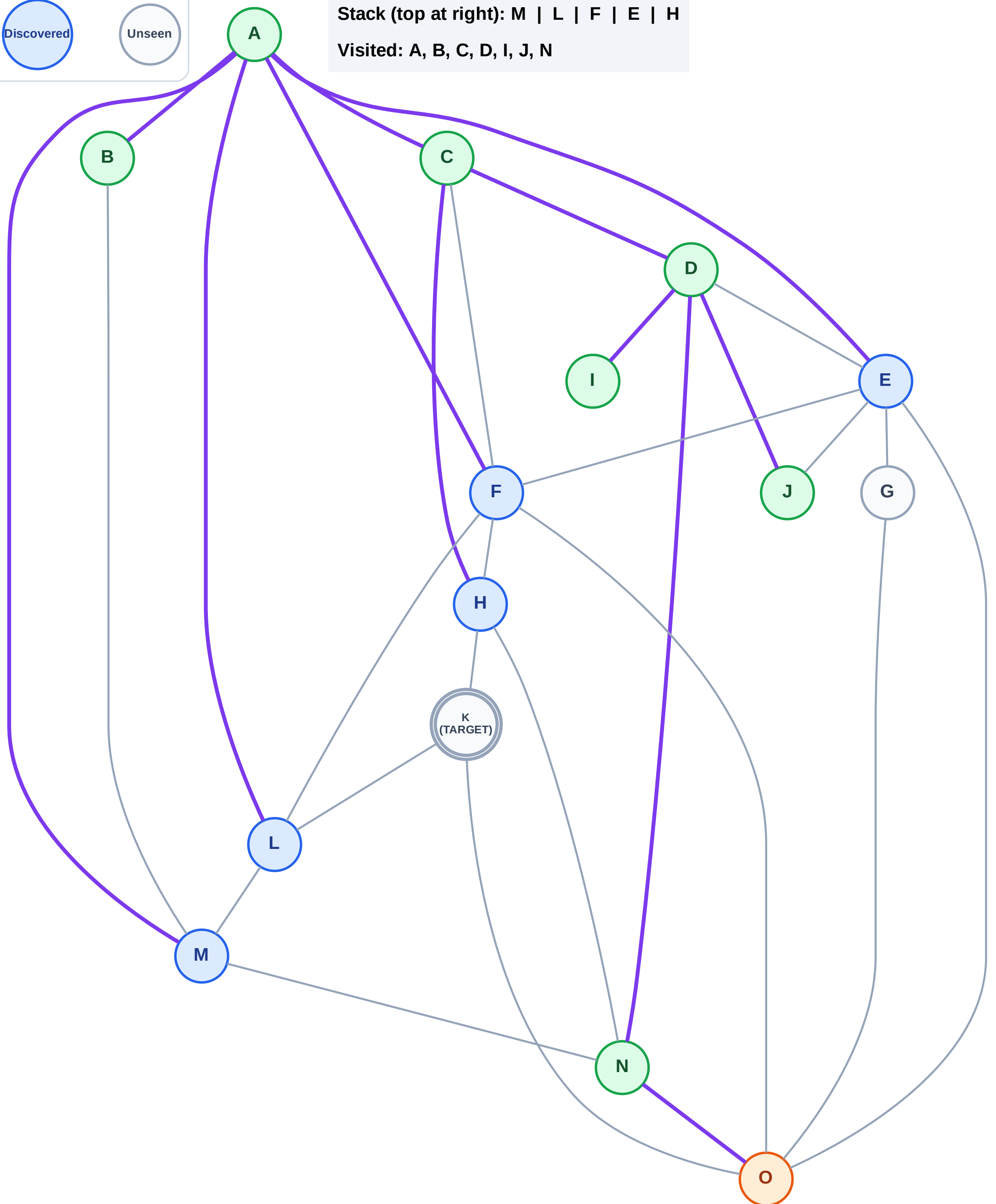
Step 16: Process node O

Legend



Stack (top at right): M | L | F | E | H

Visited: A, B, C, D, I, J, N



DFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → C → D → N → O → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | F | E | H
Visited: A, B, C, D, I, J, K, N, O

